

saturated), while the rate of passive diffusion is not rate-limited by a carrier. Another characteristic of facilitated diffusion is that the $V_{\max}$ can be increased by adding transport proteins to the membrane—this is a key regulatory aspect of the transport process.

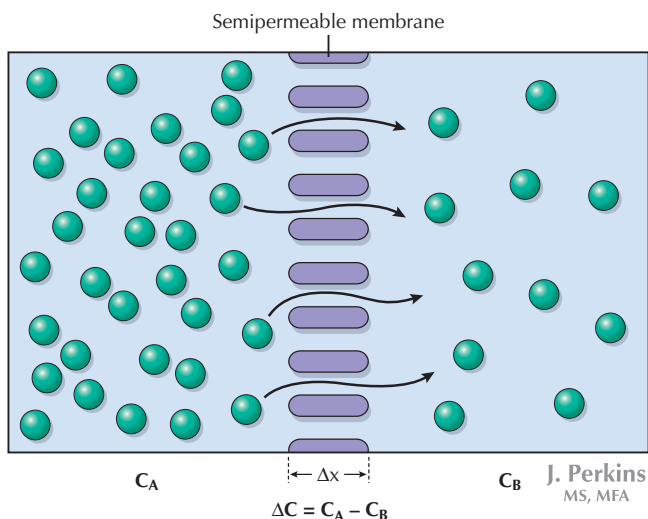


Figure 2.1 Diffusion through a Semipermeable Membrane If the membrane is permeable to a solute, diffusion can occur down the solute's concentration gradient. The rate of diffusion is contingent upon the solute gradient (ΔC) and the distance through the membrane (Δx).

Active Transport

Primary Active Transport

Primary (1°) active transport involves the direct expenditure of energy in the form of **adenosine triphosphate (ATP)** to transport an ion into or out of the cell (Fig. 2.3). Although the elements depicted in Figure 2.3 are important in many cells, the most ubiquitous is the *Na⁺ pump* (*Na⁺/K⁺ ATPase*). The Na^+ pump uses ATP to drive Na^+ out of the cells and K^+ into the cells, which establishes the essential intracellular and extracellular ion environments (Fig. 2.4). Because three molecules of Na^+ are transported out of the cell in exchange for two molecules of K^+ transported into the cell, this helps establish an electrical gradient (slightly negative inside the cell), in addition to the effects of ion diffusion due to concentration gradients (discussed in Chapter 3). The ability of the Na^+ pump to maintain the internal and external cell Na^+ and K^+ milieu is essential to cell function. If the Na^+ pump is blocked (for example, by the drug ouabain), intracellular and extracellular Na^+ and K^+ would equilibrate, affecting membrane transport and electrical potentials (see Chapter 3).

Secondary Active Transport

Many substances are transported into or out of the cell via **secondary active transport** (2° AT, also known as **cotransport**) with Na^+ . The Na^+ concentration gradient is maintained by the active Na^+/K^+ pump, which results in diffusion of Na^+ into the cell down its concentration gradient through a specific symporter or antiporter (as described earlier), allowing simultaneous transport of another molecule into or out of the

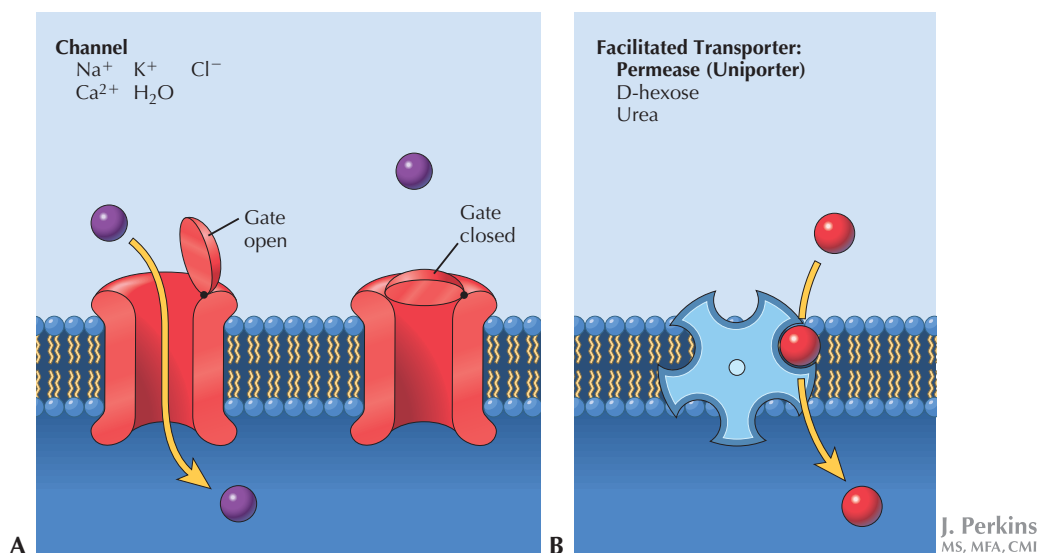


Figure 2.2 Passive Membrane Transport Passive transport of substances through the membrane can occur via specific channels or transport proteins. Channels (A) can be opened or closed depending on the position of the “gate.” The conformational change to open and close gates can be stimulated by ligand binding or changes in voltage. Specific transport proteins (B) bind substances, undergo conformational change, and release the substance on the other side of the membrane.

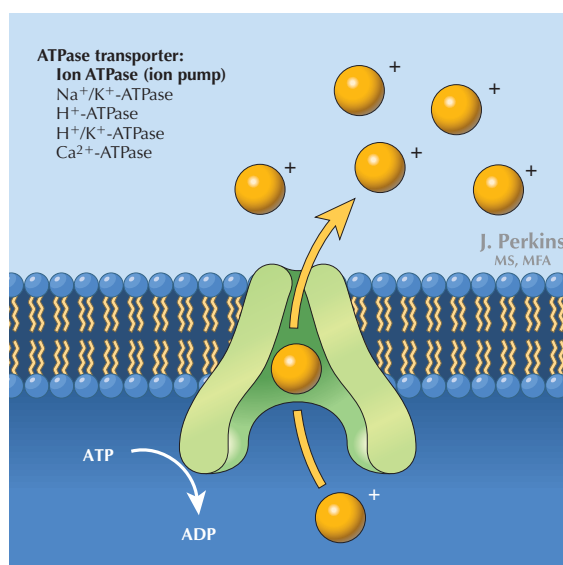


Figure 2.3 Primary (1°) Active Transport The proteins involved in primary active transport require energy in the form of ATP to transport substances against their concentration gradients.



Ouabain (or digoxin) is a cardiac glycoside derived from leaves of the foxglove plant, and the plant extract has been used in a variety of ways for hundreds of years. It is an irreversible blocker of the Na^+/K^+ ATPase pump, which will allow equilibration of sodium and potassium across the membrane, effectively stopping Na^+ -dependent transport, and depolarizing the resting membrane potential. Digoxin is used in low doses to correct cardiac arrhythmias and in congestive heart failure. The effective dose of digoxin is near the lethal dose, and excess amounts can lead to death because of the wide-ranging effects. Thus, its use is closely monitored.

cell (see Fig. 2.4). The *active* portion of this process is the original transport of Na^+ against its gradient by the Na^+/K^+ pump; the subsequent events are *secondary*. A typical example of 2° AT by symport is Na^+ -glucose and Na^+ -galactose transport across the intestinal epithelium. An example of antiport is Na^+/H^+ exchange that occurs in many cells, including renal and intestinal cells, in which Na^+ enters the cells along its concentration gradient through the antiporter while H^+ leaves the cells. The Na^+ pump also results in *passive movement* of ions through channels: Na^+ (down its concentration gradient), Cl^- (following Na^+ to preserve electroneutrality), and H_2O (following the osmotic pressure gradient) (see Fig. 2.4C).

ION CHANNELS

Movement of ions occurs through channels in addition to membrane carrier-mediated processes. **Ion channels** show high selectivity and allow specific ions to pass down their

concentration gradient (e.g., Na^+ , Cl^- , K^+ , Ca^{2+}) (see Fig. 2.2A). Selectivity depends on the size of the ion, as well as its charge. Gated channels can open or close in response to different stimuli. Stimuli such as sound, light, mechanical stretch, chemicals, and voltage changes can affect control of the ion flux by controlling the gating systems.

Types of channels include the following:

- **Ligand-gated channels** are opened by the binding of a ligand specific for that channel, such as acetylcholine (ACh). Binding of the ligand to its receptor causes the channel to open, allowing ion movement. These are tetrameric or pentameric (four or five protein subunit) channels.
- **Voltage-gated channels** open in response to a change in membrane voltage. These channels are ion-specific and are composed of several subunits, with transmembrane domains forming a pathway for ion flux across the membrane.
- **Gap junction channels** (also called hemichannels) are formed between two adjacent cells and open to allow passage of ions and small molecules between the cells. The hemichannels are generally hexameric (six subunits, or connexins).

Vesicular Membrane Transport

In addition to movement through channels and transporters, certain substances can enter or be expelled from the cell through **exocytosis**, **endocytosis**, or **transcytosis**. These types of movement through the cell membrane require ATP and involve packaging of the substances into lipid membrane vesicles for transport (Fig. 2.5).

- **Exocytosis** involves fusion of vesicles to the cell membrane for extrusion of substances contained in the vesicles.
- **Endocytosis** is the process whereby a substance or particle outside the cell is engulfed by the cell membrane, forming a vesicle within the cell. Phagocytosis is endocytosis of large particles; pinocytosis (“cell-drinking”) is endocytosis of fluid and small particles associated with the engulfed fluid.
- **Transcytosis** occurs in capillary endothelial cells and intestinal epithelial cells to move material across the cell via endocytosis and exocytosis.

Vesicular packaging and transport is especially important when the material needs to be isolated from the intracellular environment because of toxicity (antigens, waste, iron) or potential for altering signaling pathways (e.g., Ca^{2+}).

Aquaporins

In addition to ion channels, there are specific water channels or **aquaporins** that allow water to pass through the hydrophobic cell membrane, following the osmotic pressure gradient

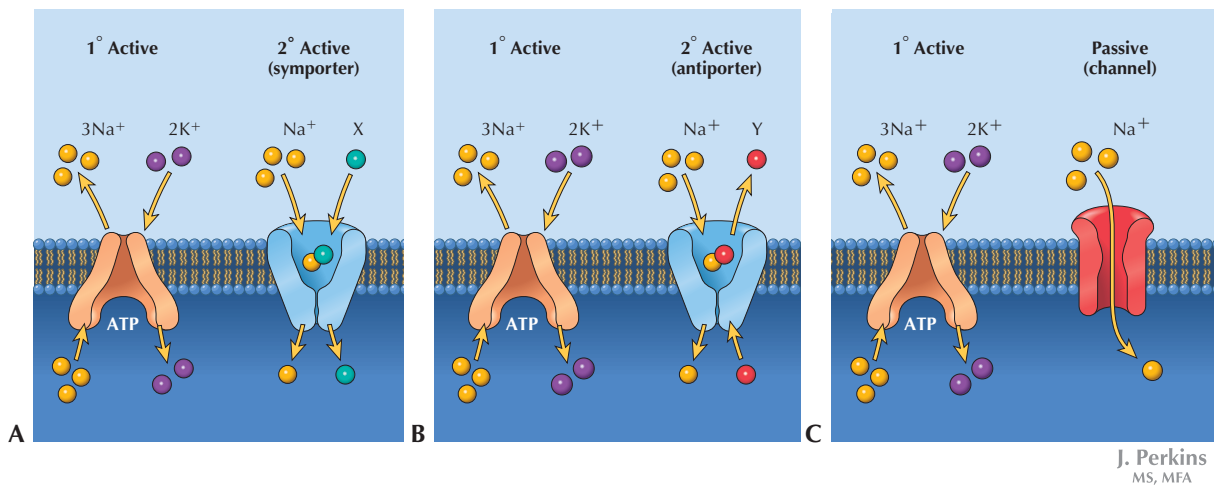


Figure 2.4 Secondary (2°) Active Transport While energy is not directly expended, secondary active transporters use a concentration gradient (usually for Na^+) established by 1° active transporters to move another substance in the same direction (symport, A), the opposite direction (antiport, B), or down the concentration gradient of the ion (C).

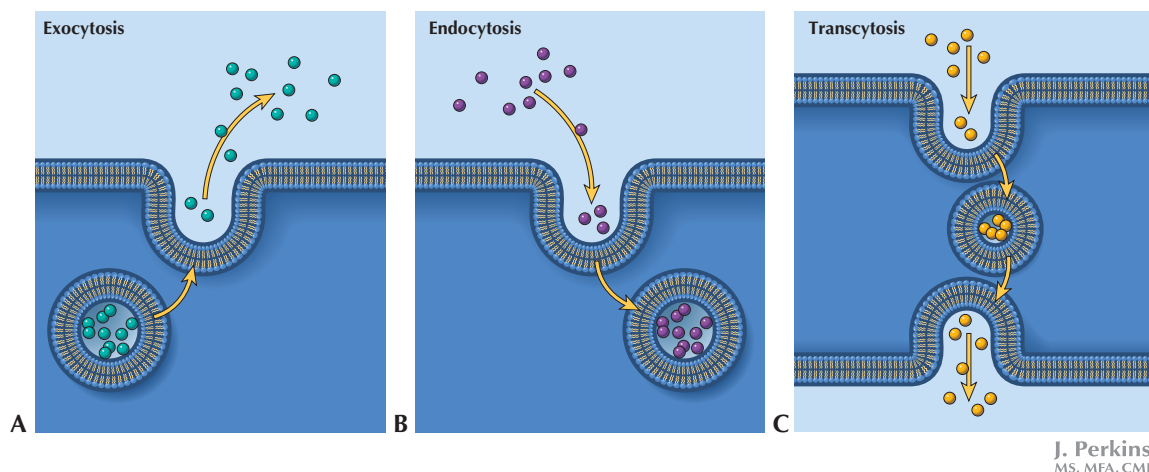


Figure 2.5 Vesicular Transport through the Membrane A, Exocytosis; B, endocytosis; C, transcytosis.

(Fig. 2.6). Many types of aquaporins (AQP) have been identified; the channels can be constitutively expressed in the membranes, or their insertion into the membrane can be regulated (for example, by antidiuretic hormone [ADH]; see Chapter 18). As exemplified in the renal cortical collecting ducts, whereas AQP-3 is always present in the basolateral membranes of principal cells, regulation of water flux is through insertion of AQP-2 into the apical (luminal) membranes.

SIGNAL TRANSDUCTION MECHANISMS

Much of the basic regulation of cellular processes (e.g., secretion of substances, contraction, relaxation, production of

enzymes, cell growth, etc.) occurs by the binding of a regulatory substance to its receptor and the coupling of the receptor to effector proteins within the cell.

Agonists, such as neurotransmitters, steroids, or peptide hormones, stimulate different transduction pathways. The pathways frequently include activation of **second messenger systems** such as cAMP, cGMP, Ca^{2+} , and IP_3 (inositol trisphosphate). The second messengers can activate protein kinases, or in the case of Ca^{2+} , calmodulin. The pathways can end in the secretion of substances, release of ions, contraction or relaxation of muscle, or regulation of transcription of specific genes, as well as other processes.

Examples of some G protein–coupled ligands using these pathways are given in Table 2.1, and some common signal transduction pathways are outlined in Table 2.2.

Protein Kinases

Many transduction pathways work through the phosphorylation of proteins by protein kinases. Protein kinase C (PK-C)

can be activated by Ca^{2+} , diacylglycerol (DAG), and certain membrane phospholipids. Protein kinases, such as PK-A, can be activated by the second messenger cAMP, and are designated “cAMP-dependent kinases.” There are also “cGMP-dependent kinases.”

Another critical pathway occurs via influx of Ca^{2+} through ligand-gated channels (Fig. 2.7), which results in activation of Ca^{2+} -calmodulin–dependent kinases. These kinases are important in smooth muscle contraction, hormone secretion, and neurotransmitter release.

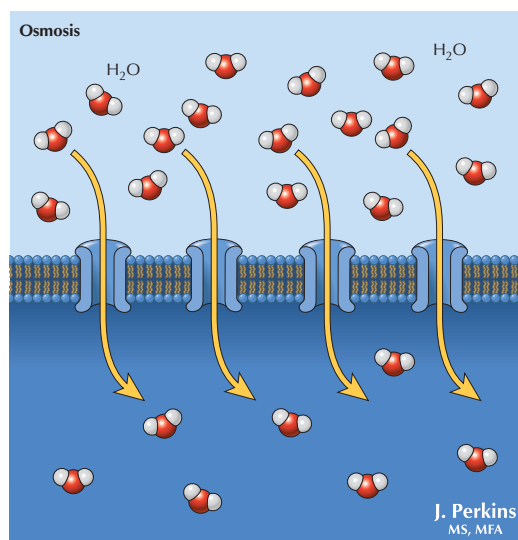


Figure 2.6 Water Channels Water flux follows the osmotic pressure gradient, and takes place through specific water channels, or aquaporins. Water movement through aquaporins can be regulated by insertion or removal of the proteins from the cell membrane.

G Proteins (Heterotrimeric GTP-Binding Proteins)

Most membrane receptors are associated with G proteins (heterotrimeric GTP-binding proteins). Ligand binding to the membrane-bound receptor-G-protein complex will cause phosphorylation of $\text{GDP} \rightarrow \text{GTP}$, allowing specific subunits of the G protein to interact with different effector proteins (Fig. 2.8). The activated G proteins also have GTPase activity, which serves to inactivate the complex and end the process. Many hormones and peptides act through this general mechanism, and examples of several G protein–coupled receptors are given in Table 2.1.

Nuclear Receptors

Several ligands, such as steroid hormones and thyroid hormone (T_3), bind directly to their **nuclear receptors**, interact with the DNA, and increase or decrease transcription of mRNA from target genes (Fig. 2.9). When this pathway is stimulated to increase protein synthesis, there is a delay in

Table 2.1 G Proteins

G Protein	Activated by Receptors for	Effectors	Signaling Pathways
G_s	Epinephrine, norepinephrine, histamine, glucagon, ACTH, luteinizing hormone, follicle-stimulating hormone, thyroid-stimulating hormone, others	Adenylyl cyclase Ca^{2+} channels	$\uparrow$ Cyclic AMP $\uparrow$ Ca^{2+} influx
G_{olf}	Odorants	Adenylyl cyclase	$\uparrow$ AMP (olfaction)
G_{r1} (rods)	Photons	Cyclic GMP phosphodiesterase	$\downarrow$ Cyclic GMP (vision)
G_{r2} (cones)	Photons	Cyclic GMP phosphodiesterase	$\downarrow$ Cyclic GMP (color vision)
$\text{G}_{\text{i1}}, \text{G}_{\text{i2}}, \text{G}_{\text{i3}}$	Norepinephrine, prostaglandins, opiates, angiotensin, many peptides	Adenylyl cyclase Phospholipase C Phospholipase A_2 K^+ channels	$\downarrow$ Cyclic AMP $\uparrow$ Inositol 1,4,5-trisphosphate, diacylglycerol, Ca^{2+} Membrane polarization
G_q	Acetylcholine, epinephrine	Phospholipase $\text{C}\beta$	$\uparrow$ Inositol 1,4,5-trisphosphate, diacylglycerol, Ca^{2+}

Note: There is more than one isoform of each class of a subunit. More than 20 distinct α subunits have been identified. ACTH, Adrenocorticotrophic hormone. (Reprinted with permission from Hansen J: Netter's Atlas of Human Physiology, Philadelphia, Elsevier, 2002.)

Table 2.2 Signal Transduction Pathways

Adenylyl Cyclase (cAMP)	Phospholipase C ($\text{IP}_3 - \text{Ca}^{2+}$)	Cytoplasmic/Nuclear Receptor	Tyrosine Kinase	Guanylate Cyclase (cGMP)
ACTH	GnRH*	Cortisol	Insulin	ANP
LH	TRH*	Estradiol	IGFs	Nitric oxide
FSH	GHRH*	Progesterone	GH	
ADH (V_2 receptor)	CRH*	Testosterone		
PTH	Angiotensin II	Aldosterone		
Calcitonin	ADH (V_1 receptor)	Calcitriol		
Glucagon	Oxytocin	Thyroid hormones		
β -Adrenergic agonists	α -Adrenergic agonists			

Summary of some hormones, neurotransmitters, and drugs and the signal transduction pathways involved in their actions on cells.

*Also increase intracellular cAMP.

ACTH, Adrenocorticotrophic hormone; ADH, antidiuretic hormone (vasopressin); ANP, atrial natriuretic peptide; CRH, corticotropin-releasing hormone; FSH, follicle-stimulating hormone; GH, growth hormone; GHRH, growth hormone-releasing hormone; GnRH, gonadotropin-releasing hormone; IGF, insulin-like growth factor; LH, luteinizing hormone; PTH, parathyroid hormone; TRH, thyrotropin-releasing hormone.

(Reprinted with permission from Hansen J: Netter's Atlas of Human Physiology, Philadelphia, Elsevier, 2002.)

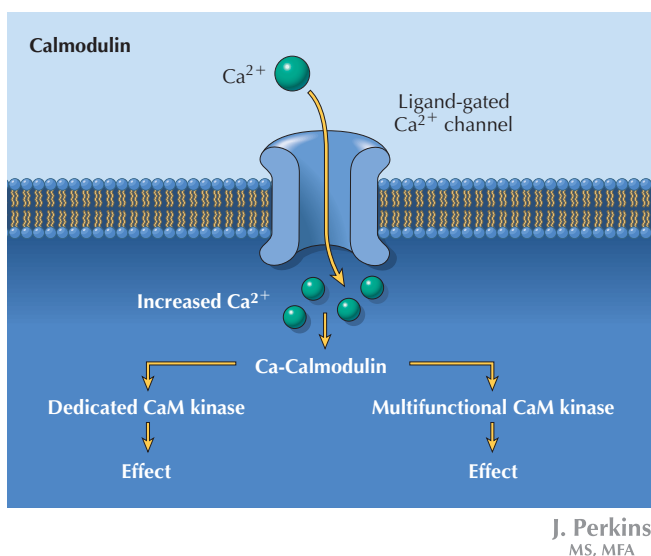


Figure 2.7 Ca^{2+} -Calmodulin Signal Transduction A good example of this mechanism is in stimulation of smooth muscle contraction, in which release of a neurotransmitter (such as tachykinin in gut smooth muscle) will act on its receptors to open Ca^{2+} channels. The increased Ca^{2+} in the cytosol binds to calmodulin, which activates specific myosin kinases. In this case, phosphorylation of myosin results in binding to actin, causing crossbridge formation and contraction of the muscle.

presentation of the end protein because the process involves gene transcription and translation. This delayed action is in contrast to that of other hormones and ligands that release proteins from storage vesicles and thus can have a rapid effect.

Simple vs. Complex Pathways

Transduction pathways can be relatively simple and fast-acting, as is usually the case for the guanylate cyclase system (cGMP). This type of rapid effect is illustrated in the smooth muscle relaxation response to nitric oxide (NO) released by vascular endothelial cells (Fig. 2.10). This is in contrast to the complex, slower transduction system observed in the multiple steps involved in growth factor signal transduction, where ligand binding initiates a multistep process ending in nuclear transcription and protein synthesis (see Fig. 2.10).

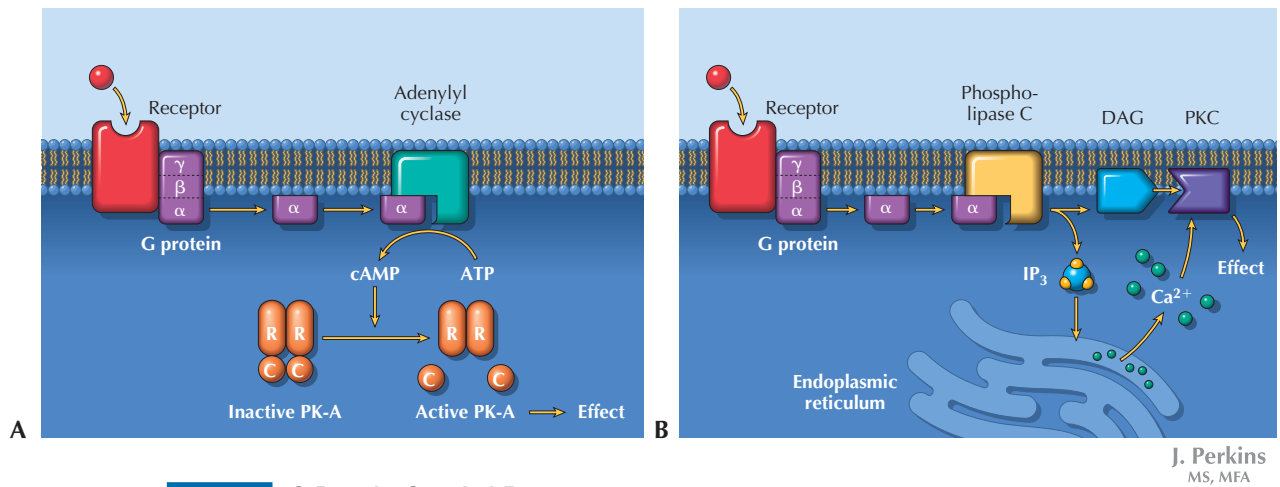


Figure 2.8 G Protein-Coupled Receptors Many ligands bind to receptors associated with membrane-bound G proteins to initiate transduction pathways. These G proteins interact with other membrane-bound proteins that activate second messenger systems. The second messengers represented (cAMP and IP_3) activate the effector proteins PK-A (A) and PK-C (B), respectively.

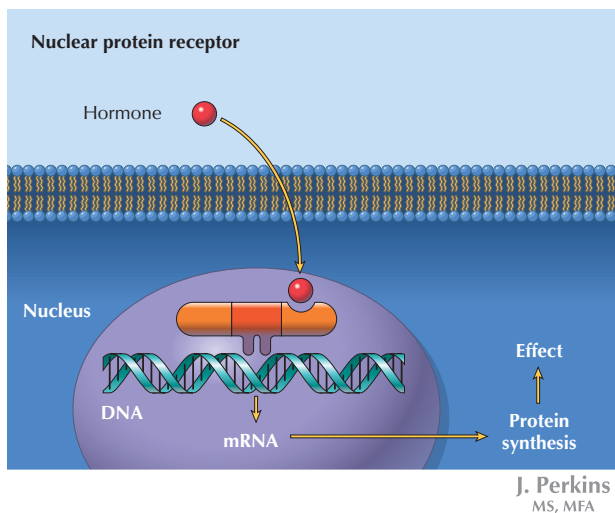


Figure 2.9 Nuclear Protein Receptors Several lipophilic hormones do not bind to the cell membrane, but instead diffuse through the membrane and are translocated through the cytosol to the nucleus. In the nucleus, they bind to receptors associated with the DNA and regulate RNA synthesis. Because this involves transcription and translation of proteins, the process takes time to produce the end effect.

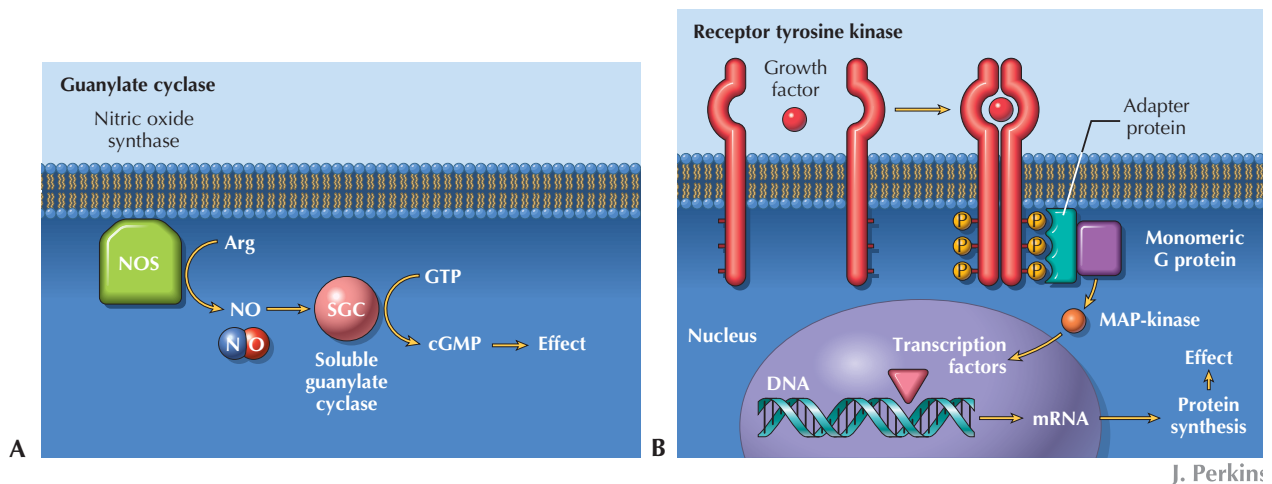


Figure 2.10 Simple and Complex Transduction Systems A, Some transduction pathways have immediate effects, such as those observed with guanylate cyclase activation of cGMP. B, Other pathways are far more complex, involving multiple effectors and nuclear transcription, which prolongs the time for the final effect.

CLINICAL CORRELATE**Cystic Fibrosis**

The importance of **membrane transporters** can be illustrated in cystic fibrosis, the most common lethal genetic disease affecting Caucasians (1 in ~2000 live births). Cystic fibrosis is caused by a defect in the *cystic fibrosis transmembrane regulator* (CFTR) gene that regulates specific apical (luminal) electrogenic chloride channels (see Fig. 2.2A). The defect has profound effects on ion and fluid transport, primarily in the lungs and pancreas. In these tissues, it is critical for Cl^- to be secreted into the lumen of the conducting airways and pancreatic acini and ducts, drawing Na^+ and water. In cystic fibrosis, the CFTR proteins are significantly reduced, decreasing Cl^- secretion and resulting in thick secretions. In the lungs, the thick, dry mucus layer contributes to increased infections; in the pancreas, the ducts from the acini are clogged with mucus and unable to secrete proper amounts of the buffers and enzymes necessary for proper digestion. The pancreatic insufficiency can result in GI complications such as meconium ileus in newborns and maldigestion, malabsorption, and weight loss as the child grows older.

Cystic fibrosis is usually diagnosed by age 2, and recently the mean age of survival was determined to be 37 years (as cited by the

Cystic Fibrosis Foundation). Though antibiotics are used to treat the frequent lung infections, there is currently no cure for the disease. Treatment includes physical therapy in which the patient's chest and back are pounded to loosen and expel the mucus. There are new methods to simulate the percussive action on the mucus, such as the intrapulmonary percussive ventilator and biphasic cuirass ventilation. As lung disease worsens, patients may need to use bilevel positive airway pressure (BiPAP) to assist in the ventilation of clogged airways.

In addition to the lung pathology, the effects on the GI tract caused by pancreatic dysfunction can necessitate surgical removal of areas of the small intestine that are amotile and usually requires supplementation of pancreatic enzymes that are reduced. In addition, reduction in endocrine pancreas function (insulin, glucagon, somatostatin) can increase the incidence of diabetes in patients with cystic fibrosis.

In all, morbidity is high and lifespan is considerably shortened. If the patient does not succumb to lung infection, the progressive decrease in lung function and exercise intolerance usually leads to lung transplant.

Clinical Features of Cystic Fibrosis

<u>Infancy</u>	Meconium ileus Meconium peritonitis Jaundice
<u>Beyond infancy</u>	Failure to thrive
<u>Childhood</u>	Steatorrhea Recurrent bronchopulmonary infections Intestinal obstructions
<u>Adult</u>	Hepatic steatosis Gallstones Biliary stricture CBD obstruction Pancreatitis

Fibrosis,
cystic dilatation
of pancreatic
acini, lamellar
secretion

Pancreas
slightly hyperemic,
granular, exaggerated
lobulation, rounded
edges

Bronchiectasis
Bronchopneumonia

Meconium
ileus

Congenital Cystic Fibrosis CBD, common bile duct.

Review Questions

CHAPTER 1: THE CELL AND FLUID HOMEOSTASIS

1. Antipyrine and inulin are injected into a 60-kg man. After equilibrium, blood is drawn, and the concentrations of the substances are determined.

	<u>Amount of indicator injected</u>	<u>Concentration in blood</u>
Antipyrine	50 mg	1.39 mg/L
Inulin	20 mg	1.67 mg/L

Using the values above, select the best answer:

- A. The intracellular fluid volume is 28 L.
- B. The interstitial fluid volume is 8 L.
- C. The plasma volume is 5 L.
- D. The total body water is 42 L.
- E. The extracellular fluid volume is 28 L.

2. Determine the pressure and direction of fluid movement (in or out of capillary) given the following Starling forces in a capillary bed where σ is approximately 1:

$$\begin{aligned} \text{HP}_c &= 30 \text{ mm Hg} & \text{HP}_i &= 3 \text{ mm Hg} \\ \pi_c &= 28 \text{ mm Hg} & \pi_i &= 8 \text{ mm Hg} \end{aligned}$$

- A. 3 mm Hg, into capillary
- B. 3 mm Hg, out of capillary
- C. 7 mm Hg, into capillary
- D. 7 mm Hg, out of capillary
- E. 19 mm Hg, out of capillary

3. Addition of pure water to the extracellular fluid (ECF) will have what effect on intracellular fluid (ICF) and ECF compartment volume and osmolarity after steady state is achieved? Assume no excretion of water, and an original plasma osmolarity of 300 mosm/L.

- A. ICF volume decreases, ECF volume increases
- B. ICF osmolarity decreases, ECF osmolarity increases
- C. ICF osmolarity increases, ECF osmolarity increases
- D. ICF volume increases, ECF volume increases
- E. ICF volume and osmolarity decrease

CHAPTER 2: MEMBRANE TRANSPORT

4. Select the **TRUE** statement about cell transport processes:

- A. Simple (passive) diffusion of a molecule is not dependent on the thickness of the cell membrane.
- B. Ion channels are relatively nonselective and allow movement of multiple electrolytes through a single channel.
- C. Voltage-gated channels are ion-specific and open in response to a change in membrane voltage.
- D. Secondary active transport directly uses ATP to move substances in and out of cells.
- E. Facilitated diffusion requires energy to move substances in and out of cells.

5. The membrane transporter directly responsible for maintenance of low intracellular sodium concentration is the:

- A. basolateral Na^+/H^+ antiporter.
- B. basolateral $\text{Na}^+/\text{Ca}^{2+}$ antiporter.
- C. apical Na^+/H^+ antiporter.
- D. basolateral Na^+/K^+ ATPase.
- E. apical Na^+/K^+ ATPase.

6. Ouabain (digoxin) works by:

- A. stimulating cGMP.
- B. blocking Na^+/K^+ ATPase.
- C. stimulating Na^+/K^+ ATPase.
- D. blocking cGMP.
- E. stimulating H^+/K^+ ATPase.

This page intentionally left blank

Section 2

THE NERVOUS SYSTEM AND MUSCLE

In complex organisms such as humans, homeostasis requires communication between distant parts of the body; continuous monitoring of changing internal and external conditions; and coordinated, often complex, responses to these conditions. This is accomplished in an integrated manner by highly differentiated tissues, organs, and systems. This demanding set of tasks is orchestrated in large part by the nervous system, along with various motor systems that receive its input.

Chapter 3 **Nerve and Muscle Physiology**

Chapter 4 **Organization and General Functions of the Nervous System**

Chapter 5 **Sensory Physiology**

Chapter 6 **The Somatic Motor System**

Chapter 7 **The Autonomic Nervous System**

Review Questions

This page intentionally left blank

Chapter 3

Nerve and Muscle Physiology

To understand the physiology of the nervous system and muscle, it is necessary to examine function in terms of relatively simple cellular processes, as well as more complicated interactions between parts of the central nervous system, peripheral nerves and receptors, and muscle. In this chapter, basic principles of neuronal and muscular function are discussed. The role of the central nervous system in integrating neural and muscular function is considered in Chapter 4.

RESTING MEMBRANE POTENTIALS

Communication through the nervous system requires generation and transmission of electrical impulses, which in turn are dependent upon the ability of cells to maintain resting membrane potentials. The term **resting membrane potential** is synonymous with **steady state potential**. A resting membrane potential is created by passive diffusion of ions through a selectively permeable membrane, producing charge separation. An example using a hypothetical cell is illustrated in Figure 3.1.

In the simplest theoretical case, if a cell membrane is permeable to only one ion and that ion is present in higher concentration inside the cell compared with outside, that ion will diffuse out of the cell until sufficient membrane potential is established to oppose further net flux of the ion. For example, if the membrane is permeable only to K^+ , and intracellular K^+ concentration is higher than extracellular concentration, then outward net flux of K^+ will occur, resulting in a negative membrane potential, in which the intracellular compartment is electrically negative relative to the outside of the cell. Only a minute fraction of the ions will diffuse out of the cell, with no appreciable change in ion concentration in the compartments, before the established electrical gradient will be sufficient to oppose further outward net flux of the ion. At this point, a resting membrane potential will be established. Due to the difference in potential between the two compartments (inside vs. outside), the excess positive charges remain close to the membrane in the extracellular compartment, while negative charges line up along the inside of the membrane.

Nernst Equation

The electrical potential difference between the inside and the outside of a cell (E_x) can be predicted if the membrane is permeable to only one ion, using the **Nernst equation** (see Fig. 3.1):

$$E_x = (RT/ZF) \ln([X]_o/[X]_i)$$

Where:

- E_x is the Nernst potential or equilibrium potential
- $\ln([X]_o/[X]_i)$ is the natural log of the ratio of the concentration of ion X outside compartment ($[X]_o$) to the concentration of the ion inside the compartment ($[X]_i$)
- R is the ideal gas constant
- T is absolute temperature
- Z is the charge of the ion
- F is Faraday's number

In biological systems at 37°C, this equation can be simplified to:

$$E_x = (61\text{mV}/Z) \log([X]_o/[X]_i)$$

Thus, in a simple hypothetical situation in which a single, monovalent cation is permeable and its concentration inside the cell is 10-fold higher than outside (see Fig. 3.1), this equation becomes:

$$E_{K^+} = (61\text{mV}) \log(0.1) = -61\text{mV}$$

$$E_{K^+} = (61\text{mV}/+1) \log(0.1\text{M}/1.0\text{M}), \text{ or } 61\text{mV}$$

The electrochemical equilibrium will be greatly changed by altering the concentration of the permeable ion inside or outside of the cell. For a system in which only one ion is permeable, the Nernst potential for the permeable ion is equal to the resting membrane potential. In actual cells, more than one ion is permeable, and thus the resting membrane potential is a result of different permeabilities (conductances) of the ions present and the concentration differences of those ions across the cell membrane. The Nernst potential (equilibrium potential) represents the *theoretical* membrane potential that would occur if the cell membrane were to be selectively permeable to only a specific ion. Given the normal values for concentrations of various ions inside cells and in the interstitial fluid, the approximate Nernst potentials for the major permeable ions are listed in Table 3.1.

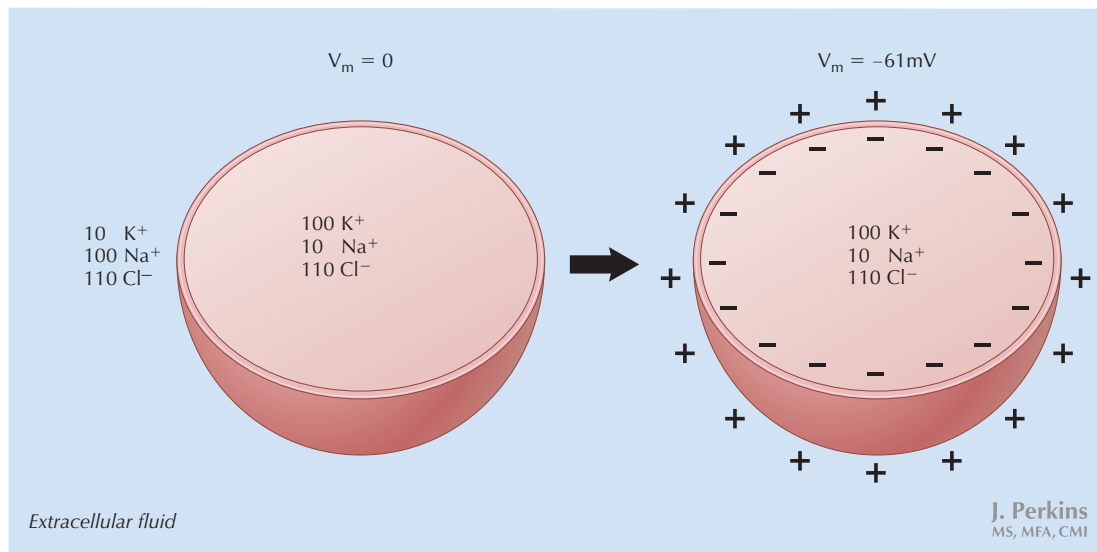


Figure 3.1 Membrane Potential of a Hypothetical Cell When the cell with the indicated intracellular ion concentrations (mEq/L) is placed into a solution with different ion concentrations, there is a gradient for diffusion of ions. In this example, the membrane separating inside and outside compartments is selectively permeable; it is permeable only to K⁺. Because the cell is selectively permeable to K⁺, diffusion of K⁺ from inside to outside the cell immediately begins, resulting in negative charge on the inside of the membrane and positive charge outside the membrane. An electrochemical equilibrium is established, in which the gradient for diffusion of K⁺ is opposed by the electrical gradient, and the membrane is negatively charged. In a simple system with only one permeable ion, V_m (the membrane potential) can be calculated by the Nernst equation.

Table 3.1 Approximate Concentrations and Nernst Potentials for Major Ions in Cytosol and Interstitial Fluid

Ion	CONCENTRATION (mM)		Nernst Potential (mV)
	Cytosolic	Interstitial	
Na ⁺	14	140	61
K ⁺	140	4	-94
Cl ⁻	8	108	-68

mM, millimolar; mV, millivolt.

- $[X]_o$ is the concentration of x outside the cell
- R is the ideal gas constant
- T is absolute temperature
- Z is the charge of the ion
- F is Faraday's number

Although cells contain many ions, this simplified G-H-K equation omits ions that are much less permeable to the cell membrane than K⁺, Na⁺, and Cl⁻, because their contribution to resting membrane potential is usually negligible. The concentration of Cl⁻ inside appears in the top of the right-hand term and the concentration of Cl⁻ outside appears in the bottom, whereas the situation for [K⁺] and [Na⁺] is opposite that of [Cl⁻], because of the difference in charge of these ions (negative vs. positive).

Goldman-Hodgkin-Katz Equation

The actual resting membrane potential (V_m) for a system involving more than one permeable ion is calculated by the Goldman-Hodgkin-Katz equation (G-H-K equation), which takes into account the permeabilities and concentrations of the multiple ions:

$$V_m = \frac{RT}{ZF} \ln \frac{P_K + [K^+]_o + P_{Na} + [Na^+]_o + P_{Cl} - [Cl^-]_i}{P_K + [K^+]_i + P_{Na} + [Na^+]_i + P_{Cl} - [Cl^-]_o}$$

Where:

- P_x is the membrane permeability to ion x
- $[X]_i$ is the concentration of x inside the cell

The resting membrane potential for most cells is approximately -70 mV; in nerve cells it is approximately -90 mV. K⁺ contributes most to the resting membrane potential, because cytosolic K⁺ concentration is high and extracellular K⁺ concentration is low, and permeability of the plasma membrane to K⁺ is high relative to other ions. Therefore, while the resting membrane potential is *similar* to the Nernst potential for K⁺, other ions contribute to the resting membrane potential. Specifically, leaking of Na⁺ through Na⁺ channels along its electrochemical gradient contributes to the fact that the resting membrane potential of cells is less negative (more positive) than the Nernst potential for K⁺. The gradients of Na⁺ and K⁺ are maintained in living cells by active transport via the

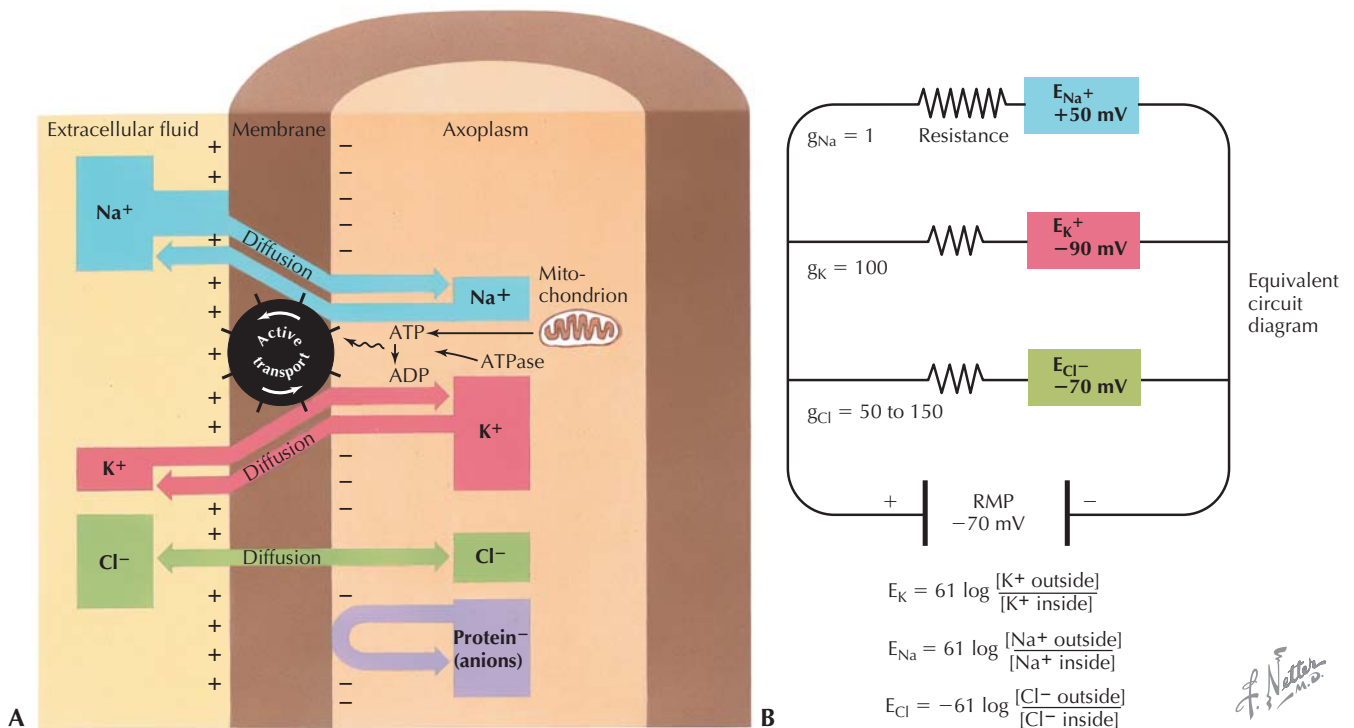


Figure 3.2 The Resting Membrane Potential Ion concentration differences between cytosol (in this example, axoplasm) and interstitial fluid are produced by the activity of transporters, mainly the Na^+/K^+ ATPase, which pumps Na^+ out of the cell in exchange for K^+ , which is pumped into the cell (A). (Size of rectangles indicates relative concentration of ions.) The cell membrane has differential permeabilities to the ions, resulting in leak across the membrane at different rates. The membrane is not permeable to proteins, which are negatively charged and are mainly intracellular. Leakage of K^+ out of the cell is the major factor in setting the resting membrane potential, because it is the most permeable ion. This situation can be modeled as an electrical circuit (B), in which potential differences for the ions are expressed (in mV), based on the values obtained from the Nernst equation, and the permeability of the ions is expressed in terms of conductance (g).

Na^+/K^+ ATPase, counteracting the constant leak of Na^+ (and thus K^+) (Fig. 3.2).

Where:

- V is the voltage of potential difference, measured in volts (V) or mV
- I is the current, measured in amperes (A), nanoamperes (10^{-9}A), or picoamperes (10^{-12}A)
- R is resistance, measured in ohms or megaohms (10^6 ohms)

In any electrical system, including movement of permeable ions across a cell membrane, current will occur (that is, charged particles will flow) driven by the potential difference in the system and impeded by the resistance of the system. Conductance (G) is the inverse of resistance and is measured in Siemens (S), nanoSiemens, or picoSiemens:

$$G = 1/R$$

In cells, conductance for various ions is a function of “leakiness” of the cell membrane to the ions (related to permeability, or presence of open channels). Ohm’s law can be restated

ELECTROPHYSICAL PRINCIPLES

Some key electrophysical principles apply to movement of ions in cellular systems, and also to conduction of electrical impulses in the nervous system. According to Ohm’s law:

$$V = IR$$



A cell is said to be hyperpolarized when its membrane potential is more negative than the normal resting membrane potential for the cell; a cell at membrane potential less negative than its usual resting membrane potential is said to be depolarized. The membrane potential is a function of the individual concentration gradients for ions across the cell membrane and the permeabilities of the membrane to those ions.

to predict a specific ion current based on membrane potential and conductance of the ion:

$$I_x = G(V_m - E_x)$$

In other words, an ion current across a cell membrane will equal the product of the conductance for the ion and the difference between the membrane potential and the equilibrium (Nernst) potential for the ion. Thus, the positive ion K^+ continues to leak out of the cell (along its concentration gradient) at resting membrane potential, because the V_m (approximately -70 mV in most cells) is less negative than E_{K^+} (-94 mV); Na^+ will continually leak into a cell, because its equilibrium potential is 61 mV compared with its resting membrane potential of -70 mV. By convention, the current associated with this outward flux of K^+ , I_{K^+} , is a positive current, whereas the inward Na^+ current, I_{Na^+} , is a negative current. Again, the concentration gradients of these ions are maintained in the face of this constant but slow leak by active transport by the Na^+/K^+ ATPase.

ACTION POTENTIALS

An **action potential** is a rapid depolarization that occurs in an excitable cell (neurons and muscle cells) initiated by an electrical event or chemical stimulation that causes increased ion permeability in the cell membrane (Fig. 3.3). Some important characteristics of action potentials are as follows:

- A **threshold potential**, which is the potential at which an action potential will be initiated by increased conductance of ions (Na^+ in many cases). For example, a cell with a resting membrane potential of -70 mV might have a threshold potential of -50 mV; if the cell is depolarized to that threshold, an action potential will occur.
- “**All-or-none**” response, with a fixed amplitude (voltage change) and stereotypic shape of the action potential for a given cell type.
- **Nondecremental propagation**, in which an action potential at one point on a cell surface will be conducted to adjacent sites along the membrane.
- A **refractory period** beginning with the initiation of the action potential, when it is impossible to elicit a second action potential.
- A **relative refractory period** following the absolute refractory period, when a second potential can be evoked but requires a larger than normal stimulus.

The threshold potential results from the balance between outward K^+ leakage and inward Na^+ current through voltage-gated Na^+ channels. When a stimulus produces depolarization that reaches this threshold, voltage-gated Na^+ channels open, and the inward flux of Na^+ now exceeds the ability of K^+ leakage to maintain a steady state. The further depolarization of the membrane opens more voltage-gated Na^+ channels, and this positive feedback continues until all voltage-gated Na^+ channels are open, producing the rapid, all-or-none depolar-

ization characteristic of an action potential. These “fast channels” are rapidly inactivated as well.

Along with these changes in Na^+ conductance, there is a delayed, slower, and smaller increase in conductance of K^+ during the action potential (see Fig. 3.3). This is caused by opening of voltage-gated K^+ channels, and along with the fall in Na^+ conductance, is responsible for the repolarization of the membrane. These K^+ channels remain open until the membrane finally returns to the equilibrium potential and are therefore responsible for the hyperpolarization or “undershoot” phase of the action potential, during which the membrane is at a more negative potential than the resting potential. This “undershoot” contributes to the relative refractoriness of the membrane to generation of another action potential, because a greater stimulus will be required to displace this hyperpolarized membrane potential to the threshold potential. A second factor contributing to refractoriness, earlier in the relative refractory period, is that some Na^+ channels still remain inactivated.

Extracellular Ca^{2+} concentration affects threshold potential by altering the sensitivity of voltage-gated Na^+ channels. At high Ca^{2+} concentrations, greater membrane depolarization is required to activate the channels, and thus the excitability of cells is reduced at high Ca^{2+} concentrations.



An **action potential** is initiated in a cell when the membrane potential is sufficiently depolarized to reach the **threshold potential**. When this occurs, opening of specific voltage-gated channels results in increased conductance of one or more ions, resulting in a rapid depolarization of the cell. In many excitable cells, the upstroke of the action potential is a result of opening of voltage-gated Na^+ channels. When the cell reaches threshold potential, a sufficient number of **activation gates** of the Na^+ channels are opened to allow Na^+ flux through the membrane, which causes the upstroke of the action potential. Shortly thereafter, **inactivation gates** close the channels, terminating the upstroke. These inactivation gates remain open until the cell returns to the resting membrane potential. Influx of K^+ is responsible for repolarization of the membrane.

ACTION POTENTIAL CONDUCTION

In excitable cells, an action potential is rapidly propagated over the cell membrane. In a neuron, when **threshold potential** is reached, an action potential is generated at the axon hillock, spreads to the axon, and is conducted to the axon terminal (Fig. 3.4) (conduction *between* cells is discussed later). As depolarization occurs at a point on the membrane, Na^+ flows through the membrane from the extracellular fluid. Loss of positive charge at the point of depolarization causes **local currents**, in which positive charges flow from adjacent regions along the membrane to the area of depolarization (Fig. 3.5A). Local current results in depolarization of the adjacent regions;

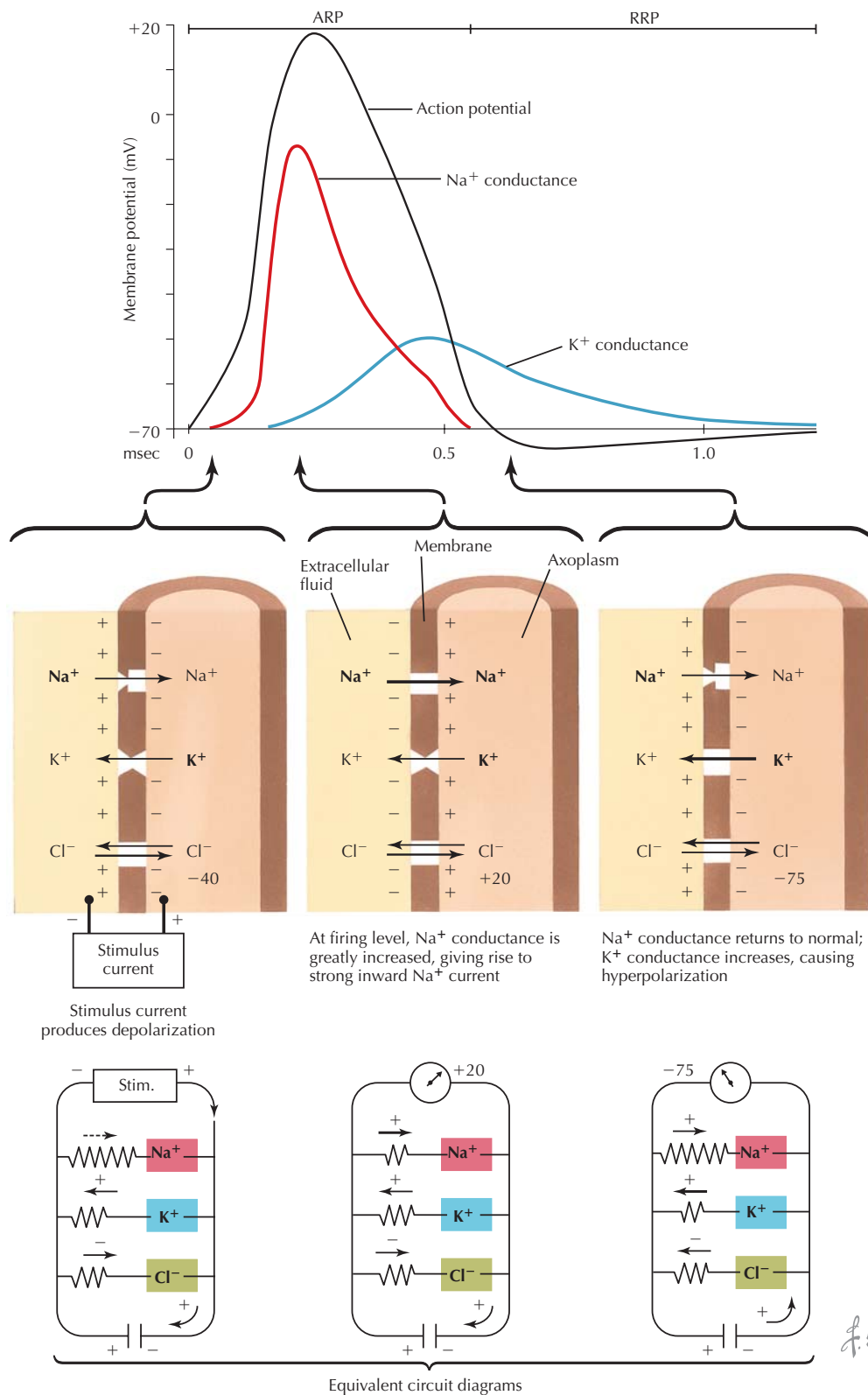


Figure 3.3 Action Potential The action potential in most excitable cells is initiated when a local potential on a cell reaches a threshold at which voltage-gated Na^+ channels open, greatly increasing Na^+ conductance. Changes in Na^+ and K^+ conductance during the action potential are illustrated, along with the equivalent circuit diagrams.

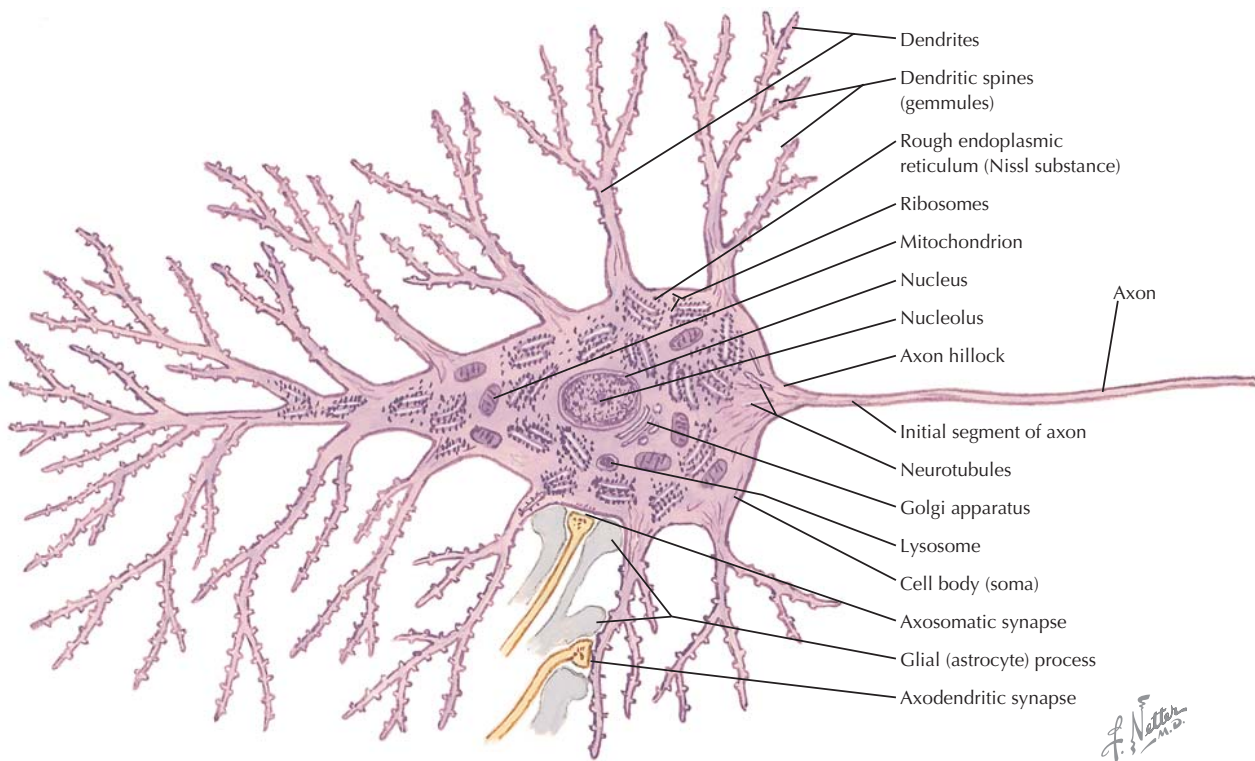


Figure 3.4 Neurons Neurons consist of the soma (cell body), multiple dendrites, and an axon. Neurons receive signals from other neurons at synapses, resulting in changes in local membrane potential. When the membrane potential is sufficiently depolarized, an action potential is initiated at the axon hillock and is propagated down the length of the axon to synapses with other neurons or muscle cells.

when threshold is reached, the action potential is propagated. An important characteristic of **action potential propagation** is that it occurs away from the point of initiation; it cannot travel back toward its origin. As the action potential is conducted, the area of the membrane directly behind the action potential is still in the absolute refractory state due to Na^+ channel inactivation, preventing retrograde conduction.

An essential property of neurons is the ability to rapidly propagate an action potential, analogous to an electrical cable. Conduction velocity is a function of:

- **Internal resistance (R_i)**, the impedance to current flow within the cytoplasm (for example, resistance within the axon).
- **Membrane resistance (R_m)**, the impedance to current flow through the membrane.
- **Capacitance (C_m)**, the ability of the membrane to store charge.

The relationships between these variables result in the **cable properties** of cells: the **space constant (λ)** and the **time constant (τ)**, which are useful in describing conduction of electrical activity in an axon. The time it takes for membrane

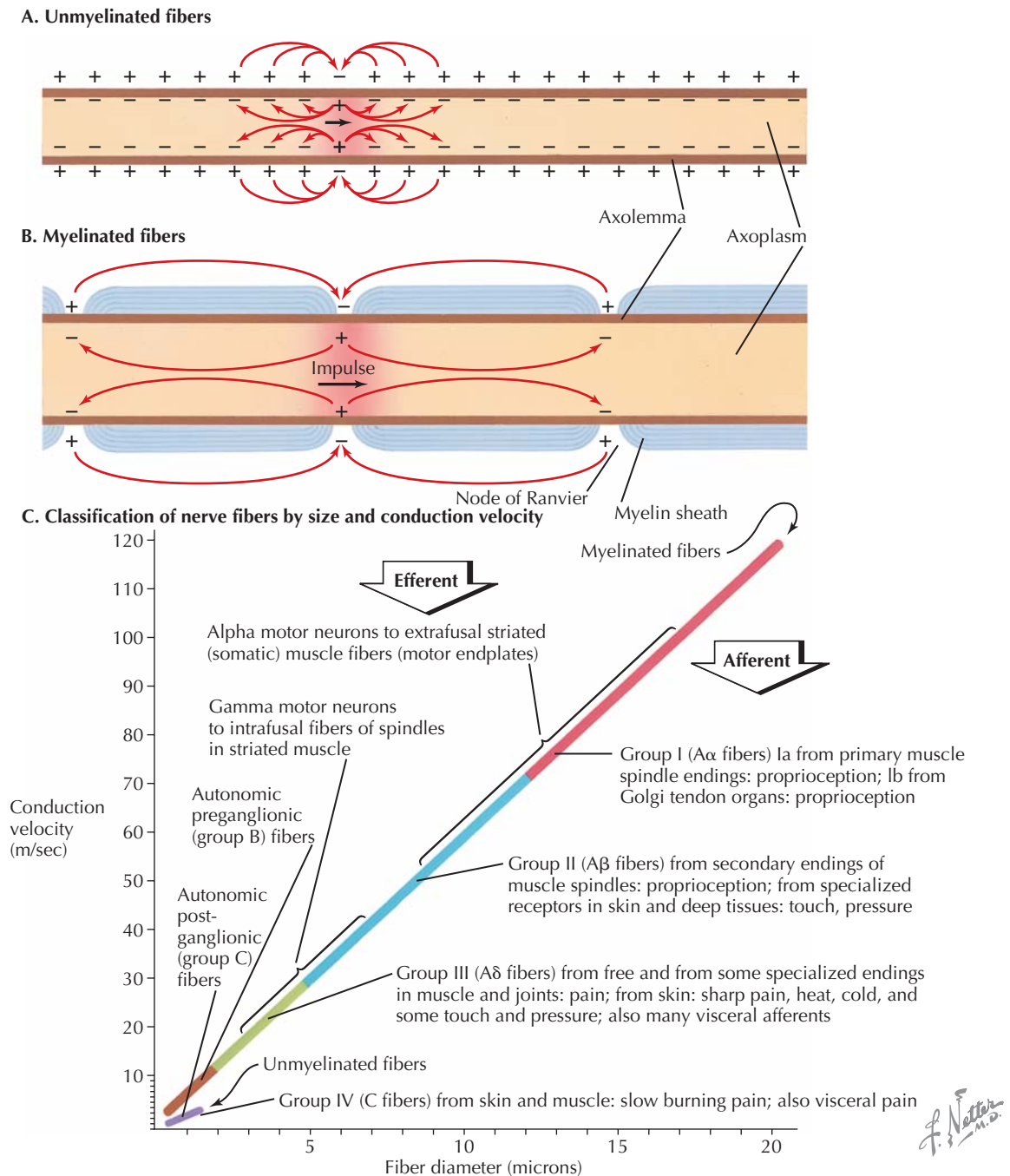
potential to change to 63% of its final value when a current is applied is τ ($\tau = R_m \times C_m$). It is a measure of how slowly or quickly the membrane depolarizes or repolarizes, and is therefore inversely related to action potential conduction velocity. Because τ is inversely related to conduction velocity, low membrane resistance and low membrane capacitance favor faster conduction.

The space constant (λ , also known as the length constant) is the distance along the membrane over which a potential change will fall to 63% of its value, and therefore describes how far a depolarizing current will spread. With greater λ , depolarizing currents will spread further, resulting in greater conduction velocity. λ is proportional to the ratio of R_m to R_i :

$$\lambda = \sqrt{(R_m/R_i)}$$



The importance of internal resistance (R_i) in action potential conduction velocity can be inferred from the anatomy of axons of large invertebrates. The giant axon of the squid has a diameter of 500 μm ; with this increased diameter, λ is high because R_i is low, resulting in the rapid conduction necessary in a large animal.



The nervous system of vertebrates is much more complex and composed of a much larger number of neurons than that of invertebrates. Many nerve cells in the vertebrate nervous system are **myelinated** (Fig. 3.5B). They are covered by multiple layers of an insulating sheath of phospholipid membrane, formed by Schwann cells in the peripheral nervous system and oligodendrocytes in the central nervous system. This insulation greatly decreases capacitance and also increases membrane resistance, such that current travels through the interior of the axon, but not across the membrane. To allow propagation of the action potential, there are breaks in the myelin sheath called **nodes of Ranvier**. They are present at 1 to 2 mm intervals along the axon, and the action potential thus “jumps” rapidly from node to node, bypassing

the myelinated areas. Essentially, the nodes of Ranvier represent multiple capacitors in series. This process of conduction whereby the action potential “jumps” between nodes is known as **saltatory conduction** and allows very rapid propagation of the action potential, despite small axon diameter. Local currents generated at one node during depolarization result in depolarization at the next node, and the action potential skips along the axon, bypassing the highly insulated segments. Fibers throughout the nervous system vary in terms of diameter and conduction velocity and may be classified into various types on this basis (Fig. 3.5C). In contrast to saltatory conduction, in unmyelinated axons, action potentials are conducted as slower waves of depolarization as they are propagated along the axon.

CLINICAL CORRELATE

Multiple Sclerosis

Multiple sclerosis (MS) is an inflammatory neurodegenerative disease, believed to be of autoimmune etiology. It is more common in women than men, and the usual age of onset is between 20 and 40 years of age. The inflammatory process of MS results in the gradual destruction of myelin sheaths around myelinated axons of the brain and spinal cord. A wide variety of symptoms may occur, including muscle weakness and paralysis; impaired coordi-

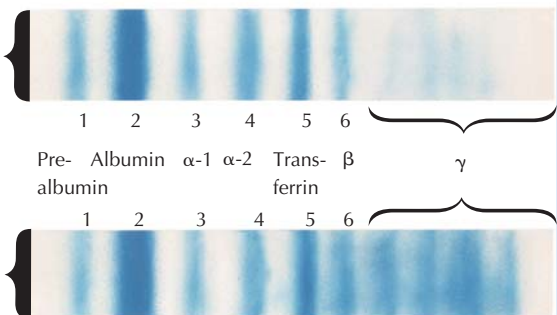
nation and poor balance; depression; impaired speech; memory problems; visual problems; altered sensory perception; pain; fatigue; and bowel, bladder, and sexual dysfunctions, resulting from the deficits in neural conduction caused by damage to myelin sheaths and underlying axons in the central nervous system. Diagnosis is based on clinical findings, MRI evidence of demyelinating lesions of the brain and spinal cord, and characteristic oligoclonal bands of γ -globulins in cerebrospinal fluid (CSF).

Cerebrospinal fluid electrophoresis



F. Netter M.D.

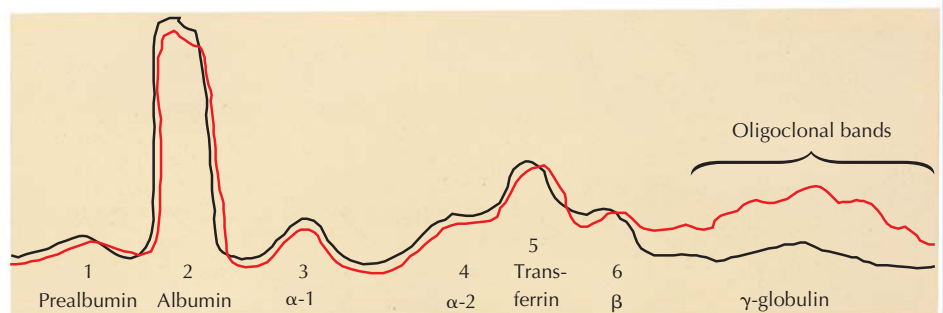
Normal



Multiple sclerosis:
 γ -globulin elevated

Computed recordings

— Normal
— Multiple sclerosis



Multiple Sclerosis: Diagnostic Tests—Spinal Fluid CSF samples obtained by lumbar puncture display characteristic elevation of γ -globulins (oligoclonal bands) upon gel electrophoresis in approximately 90% of patients.

SYNAPTIC TRANSMISSION

A **synapse** is a site at which an electrical response in one cell is transmitted to another cell. This can occur between some cell types through **electrical synapses**, when electrical current passes directly from one cell to another through **gap junctions**. Cardiac myocytes and some types of smooth muscle communicate through this mechanism (some neurons communicate both through electrical and chemical transmission). At electrical synapses between cells, intramembrane proteins called **connexons** form channels that allow flow of ions between cells, thus allowing current transmission.

At a **chemical synapse**, release of a neurotransmitter by a neuron results in electrical stimulation of the presynaptic cell. A wide variety of neurotransmitters are found in various regions of the central and peripheral nervous system (Table 3.2). The morphology of a chemical synapse is illustrated in Figure 3.6. The sequence of events in transmission at this type of synapse is:

1. Depolarization of the neuron reaches the **synaptic bouton** (axon terminal), where it causes opening of Ca^{2+} channels.
2. The influx of Ca^{2+} results in the release of neurotransmitter, stored in presynaptic vesicles, into the **synaptic cleft** (the gap between the cells).
3. The transmitter diffuses across the synaptic cleft and binds to specific membrane receptors on the postsynaptic membrane.

4. Binding of the neurotransmitter produces a change in the membrane potential of the postsynaptic membrane.

Depending on the presynaptic neuron involved (and thus, the specific neurotransmitter released), the chemical transmission may result in either an **excitatory postsynaptic potential** (EPSP; depolarization) or an **inhibitory postsynaptic potential** (IPSP; hyperpolarization), as a result of Na^+ or Cl^- influx at the postsynaptic membrane, respectively (Fig. 3.7). Chemical transmission of this type is **unidirectional** between a presynaptic fiber and a postsynaptic cell. A given neuron typically receives synaptic input from multiple inhibitory and excitatory neurons; therefore, generation of an action potential depends on summation of these multiple inputs. An action potential may occur as a result of **temporal summation** when a series of impulses is generated by an excitatory presynaptic fiber, or as a result of **spatial summation** of local potentials generated by multiple excitatory fibers in the postsynaptic cell (Fig. 3.8). Synaptic inhibition may be either presynaptic or postsynaptic; some inhibitory fibers act directly on an excitatory fiber (**presynaptic inhibition**), while others produce **postsynaptic inhibition** (Fig. 3.9). Summation of local potentials in the postsynaptic membrane is dependent on the space constant and time constant of the membrane.

NEUROMUSCULAR JUNCTION

Motor neurons (motoneurons) are efferent nerves that originate in the central nervous system and communicate with

Table 3.2 Summary of Some Neurotransmitters and Where within the Central and Peripheral Nervous System They Are Found

Transmitter	Location	Transmitter	Location
Acetylcholine	Neuromuscular junction, autonomic endings and ganglia, CNS	Gas Nitric oxide	CNS, GI tract
Biogenic amines		Peptides	
Norepinephrine	Sympathetic endings, CNS	β -Endorphins	CNS, GI tract
Dopamine	CNS	Enkephalins	CNS
Serotonin	CNS, GI tract	Antidiuretic hormone	CNS (hypothalamus/posterior pituitary)
Amino acids		Pituitary-releasing hormones	CNS (hypothalamus/anterior pituitary)
γ -Aminobutyric acid (GABA)	CNS	Somatostatin	CNS, GI tract
Glutamate	CNS	Neuropeptide Y	CNS
Purines		Vasoactive intestinal peptide	CNS, GI tract
Adenosine	CNS		
Adenosine triphosphate (ATP)	CNS		

CNS, Central nervous system; GI, gastrointestinal.

(Reprinted with permission from Hansen J: Netter's Atlas of Human Physiology, Philadelphia, Elsevier, 2002.)

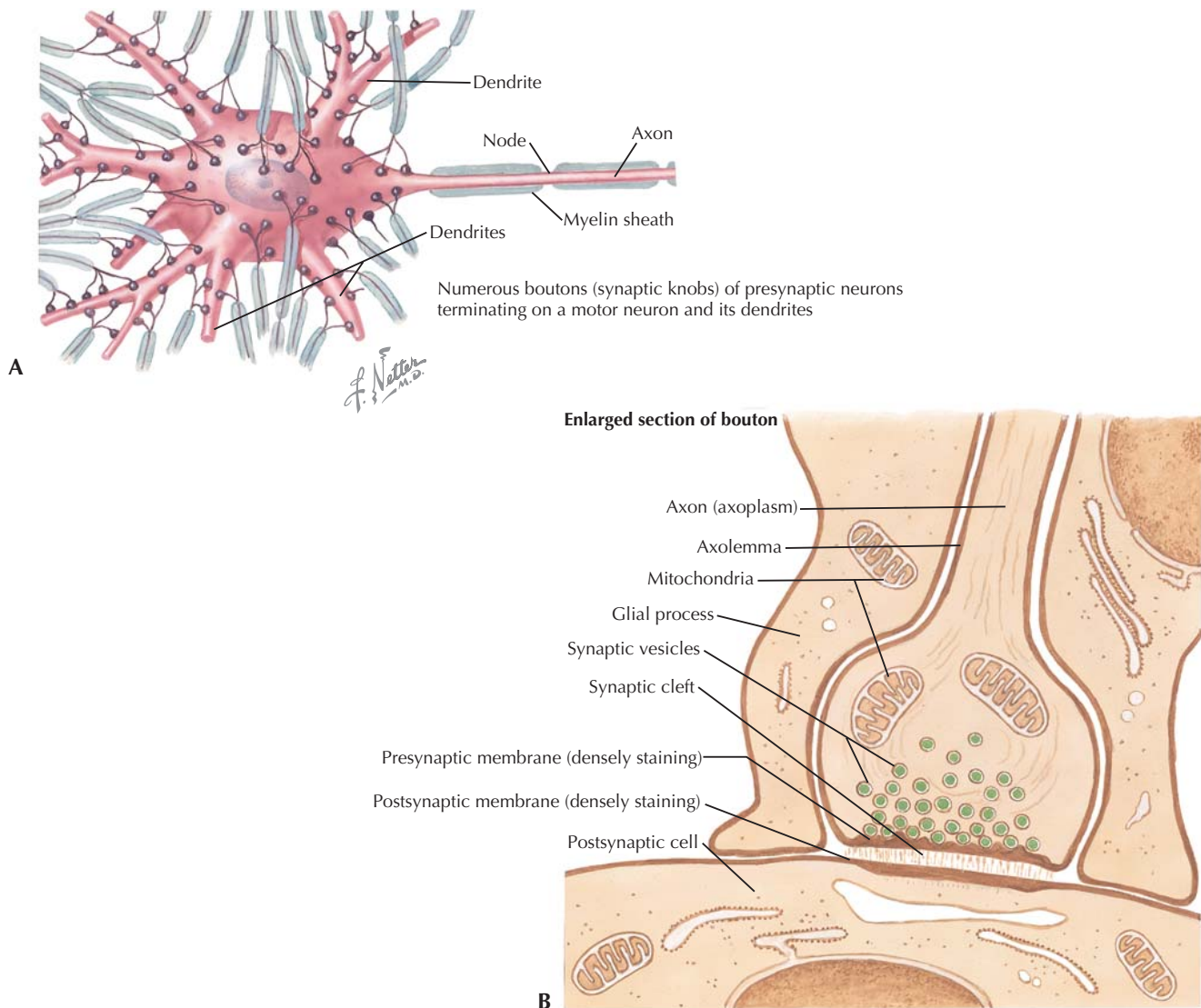


Figure 3.6 Morphology of Synapses An action potential in one neuron is transmitted to another neuron or an effector tissue at the synapses. The terminal regions of axons branch and form synaptic boutons. The myelination is lost where the bouton is in close contact with a dendrite or cell body of another neuron (A). Neurotransmitters are released into the synaptic cleft when vesicles of the synaptic bouton fuse with the presynaptic membrane (B). The released neurotransmitters diffuse and bind to receptors on the postsynaptic membrane, resulting in excitatory or inhibitory effects at the postsynaptic membrane.

skeletal muscle fibers at specialized synapses known as **motor endplates** or **neuromuscular junctions** (Fig. 3.10). Among the several types of motor neurons, the most common is the α -motor neuron. Branches of an α -motor neuron may form multiple neuromuscular junctions in depressions in the **sarcolemmae** (muscle cell membranes) of muscle fibers; each α -motor neuron may thus innervate multiple muscle fibers, although each fiber is innervated by only one α -motor neuron. An α -motor neuron and the fibers it innervates is called a **motor unit**. All the motor neurons innervating a muscle are collectively known as a **motor neuron pool**.

The synaptic boutons of α -motor neurons are rich in vesicles containing the neurotransmitter acetylcholine. As is the case for other neurons, an action potential in an α -motor neuron produces Ca^{2+} influx at the synaptic bouton, which results in release of neurotransmitter from vesicles of the synaptic bouton. Vesicular release of neurotransmitter is **quantal** in nature, and is conceptualized as the release of packets (quanta) of neurotransmitter, the quanta being defined by the response to the neurotransmitter contained in a single vesicle. At the postsynaptic membrane of a neuromuscular junction, acetylcholine binds to a specific subtype of acetylcholine receptor known as the nicotinic

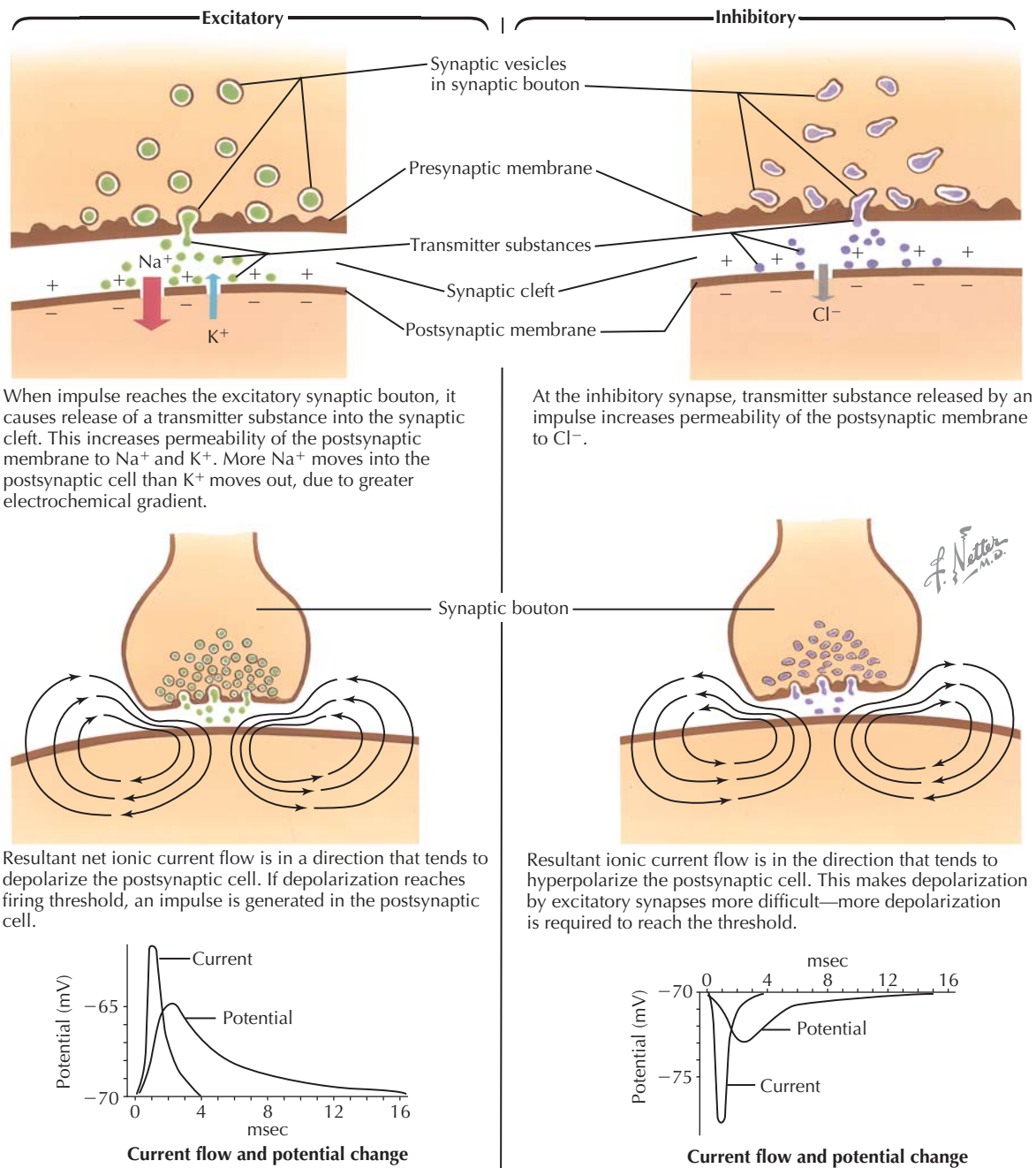


Figure 3.7 Chemical Synaptic Transmission Synaptic transmission may result in either excitatory or inhibitory effects at the postsynaptic membrane. In excitatory transmission (*left*), a depolarizing change in local potential (excitatory postsynaptic potential; EPSP) is produced. In inhibitory transmission (*right*), a hyperpolarizing change in local potential (inhibitory postsynaptic potential; IPSP) is produced. An action potential occurs when the temporal and spatial summation of local potentials reaches the threshold for action potential generation (see Fig. 3.8).

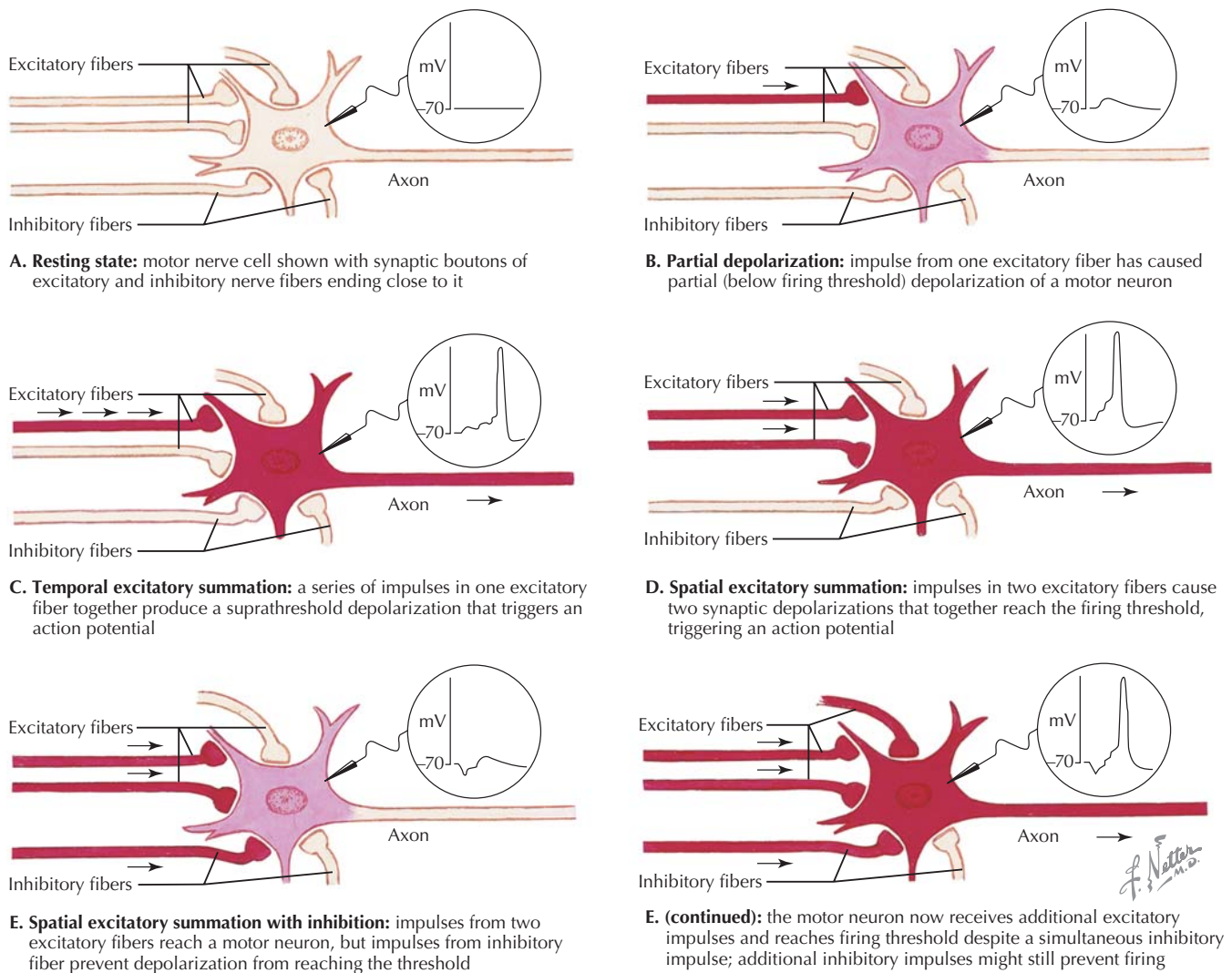


Figure 3.8 Temporal and Spatial Summation In the resting state (A), the nerve cell is at its resting membrane potential. Neurons are subject to multiple inhibitory and stimulatory signals that produce local potential changes, which are subject to both temporal and spatial summation. The sum of these influences determines whether an action potential is produced (B–E).

receptor. The action of acetylcholine is rapid; it is also short-lived because acetylcholine diffuses out of the synaptic cleft or is rapidly broken down by acetylcholinesterase localized on the basement membrane of the muscle fiber.

Binding of postsynaptic nicotinic receptors by acetylcholine causes the opening of ligand-gated channels. Influx of Na^+ and K^+ through these channels produces an excitatory postsynaptic potential (endplate potential). When this endplate potential reaches threshold, an action potential is generated, ultimately producing contraction of the muscle fiber. Pharmacologically, the actions of acetylcholine can be blocked by **curare**, a competitive antagonist that binds reversibly to the nicotinic receptor, thereby blocking acetylcholine binding, or

by **cobra venom (α -bungarotoxin)**, a noncompetitive antagonist that binds irreversibly to the receptor.

Endplate potentials differ from neuronal action potentials in several important ways:

- Endplate potentials are produced by a ligand-gated channel; neuronal action potentials are caused by voltage-gated channels. In both cases, the action potential is generated in the postsynaptic cell when a threshold is reached.
- Rapid depolarization, to a potential of 0 mV, compared with rapid depolarization to +40 mV during a neuronal action potential.

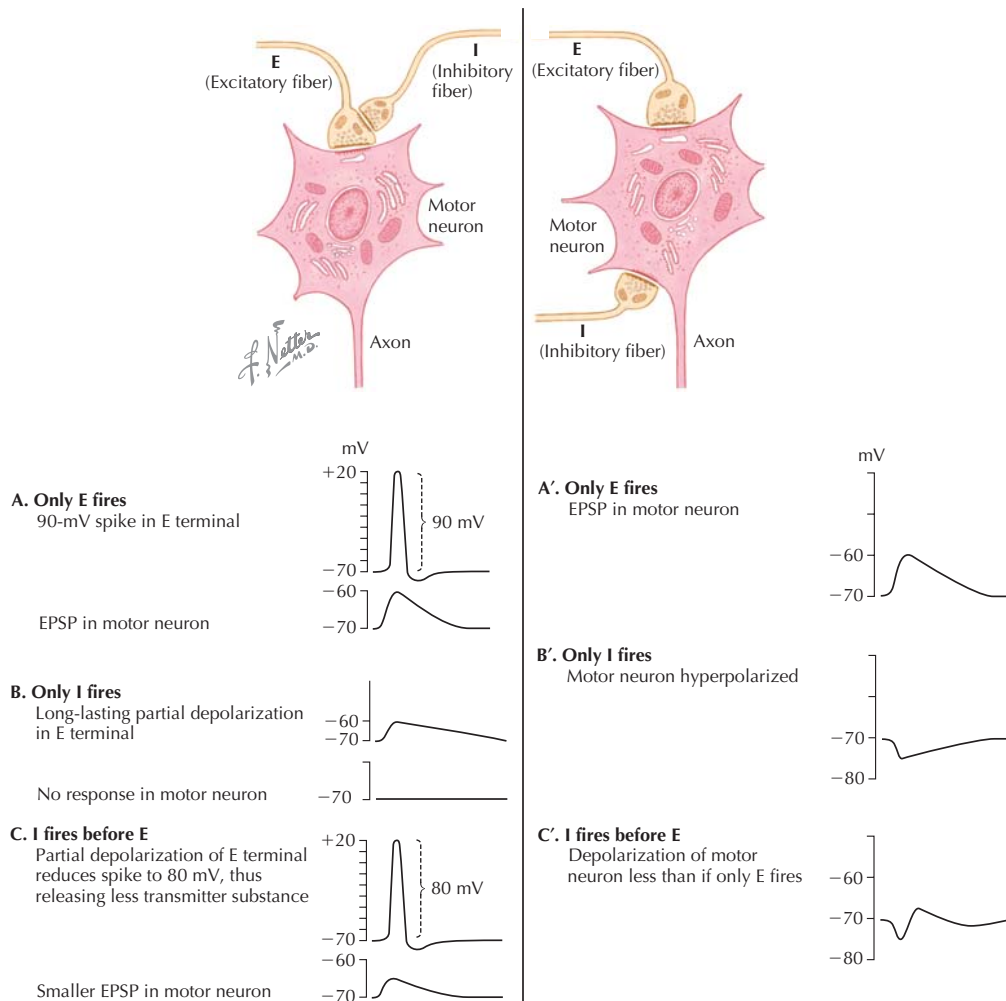


Figure 3.9 Synaptic Inhibitory Mechanisms Inhibition can occur through either presynaptic or postsynaptic mechanisms. In presynaptic inhibition, illustrated (*top left*) for a motor neuron, an inhibitory fiber of one nerve cell has a synapse on an excitatory axon of another neuron before the latter communicates with the motor neuron; in postsynaptic inhibition (*top left*), the inhibitory and excitatory fibers both synapse directly with the target neuron. A, B, and C illustrate the changes in potential at the excitatory terminal and in the motor neuron when only the excitatory, only the inhibitory, or both inhibitory and excitatory fibers fire. A', B', and C' illustrate membrane potential changes in the motor neuron when only excitatory, only inhibitory, or both excitatory and inhibitory fibers fire.

- A single, large channel for Na^+ and K^+ carries the charge during an endplate potential. Multiple ion channels are involved in a neuronal action potential, which is mainly produced by Na^+ influx.
- Repolarization of the endplate is passive, whereas increased K^+ conductance is responsible for repolarization during a neuronal action potential.

SKELETAL MUSCLE ORGANIZATION

Skeletal muscle contraction is the basis for voluntary muscle movement. Multinucleated cells contain **sarcomeres**, specialized structures that produce contraction upon stimulation of

the muscle. Sarcomeres are contained within **myofibrils**, which are further organized as **muscle fibers**; groups of muscle fibers form **muscle fascicles** (Fig. 3.11). Microscopically, the arrangement of sarcomeres and myofibrils results in the striated appearance of skeletal muscle. Contraction is based on sliding of the thin and thick filaments of sarcomeres (see “Excitation-Contraction Coupling”).

Within the skeletal muscle cell, the **sarcoplasmic reticulum** composes a complex network surrounding the myofibrils (Fig. 3.12). This specialized form of smooth endoplasmic reticulum is the site for the storage of high concentrations of intracellular Ca^{2+} and contains Ca^{2+} -ATPase and calsequestrin (a low-affinity Ca^{2+} -binding protein) for sequestration of this ion, as

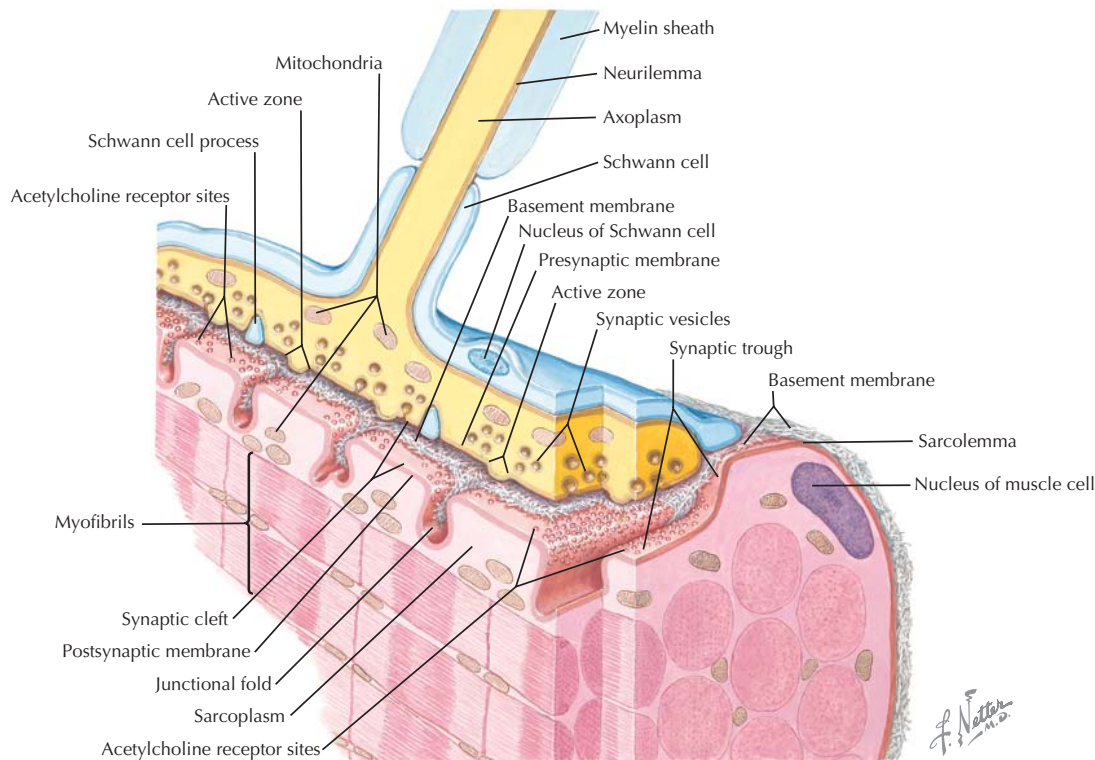


Figure 3.10 Structure of the Neuromuscular Junction At the neuromuscular junction, the axon of a motor nerve synapses with skeletal muscle at a site known as the motor endplate. Stimulation of a motor nerve results in the release of acetylcholine from vesicles at the presynaptic membrane; acetylcholine diffuses and binds to postsynaptic receptors, producing depolarization of the sarcolemma and leading to an action potential.

well as L-type (ligand-gated) Ca^{2+} channels. The **transverse tubules (T-tubules)** are deep invaginations of the muscle cell membrane (sarcolemma). They form **triads** with two terminal cisternae of the sarcoplasmic reticulum; these triads are arranged perpendicularly to the muscle fiber. The T-tubules extend into the muscle fiber from the surface, allowing close communication between the interior of the cell and the extracellular fluid. They are responsible for conducting the action potential to the cisternae of the sarcoplasmic reticulum.

EXCITATION-CONTRACTION COUPLING

As depolarization spreads across the sarcoplasmic reticulum, it is conducted into the transverse tubules (Fig. 3.13). The T-tubule membrane contains voltage-gated Ca^{2+} channels, also known as **dihydropyridine receptors**. Although the dihydropyridine receptor is a voltage-gated Ca^{2+} channel, ion flux through this channel is not required for contraction of skeletal muscle. Rather, a conformational change in the dihydropyridine receptor, caused by depolarization of the T-tubule, is required. These receptors are in close apposition to calcium channel proteins known as **ryanodine receptors**, which are large proteins of the sarcoplasmic reticulum that extend into

the gap between the cisternae of the sarcoplasmic reticulum and the T-tubules. Conformational change of the dihydropyridine receptors is believed to produce a subsequent conformational change in the ryanodine receptors, allowing stored Ca^{2+} to be released from the sarcoplasmic reticulum, initiating the contraction process. The term **excitation-contraction coupling** refers to this linking of depolarization to Ca^{2+} release.

To summarize these events:

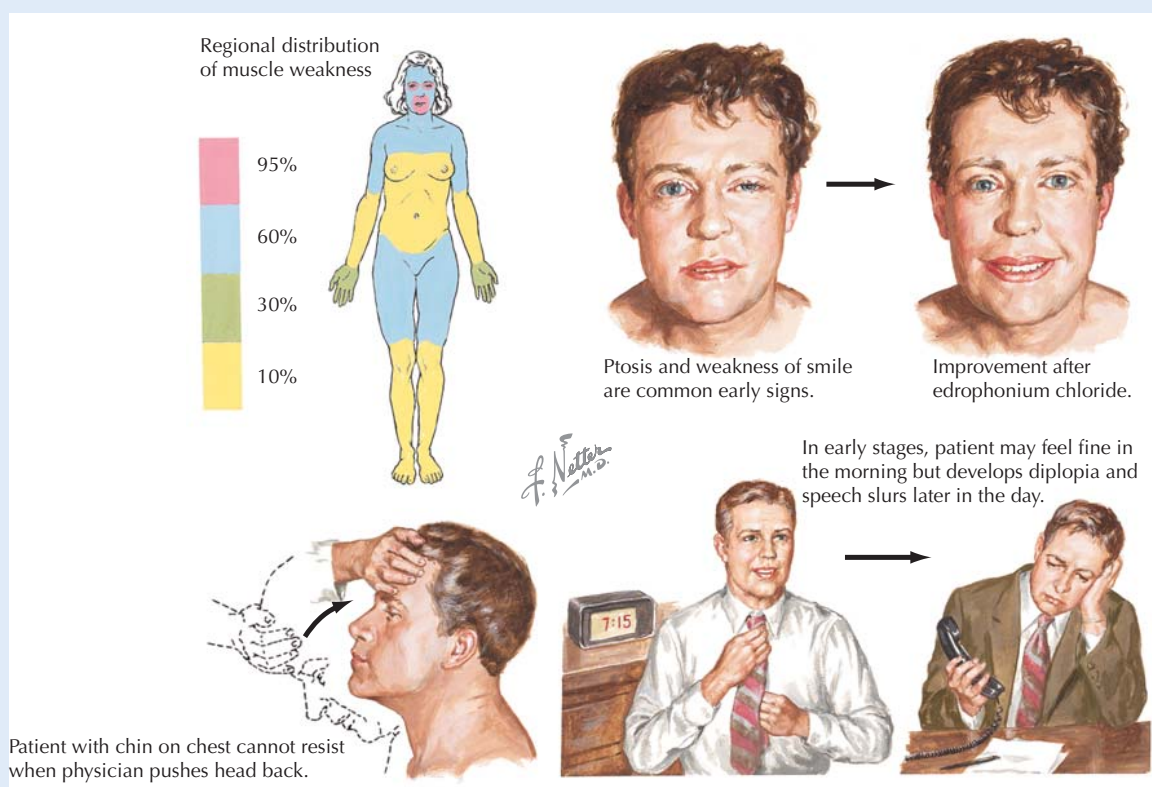
- Depolarization of the motor neuron terminal results in Ca^{2+} influx.
- Vesicles of the axon terminal release acetylcholine.
- Binding of acetylcholine by nicotinic receptors results in an endplate potential.
- An action potential is initiated and is propagated along the sarcolemma and down the T-tubules.
- A conformational change in the dihydropyridine receptor of the T-tubule is transduced to a conformational change in the ryanodine receptor of the sarcoplasmic reticulum.
- Ca^{2+} is released from the sarcoplasmic reticulum, initiating contraction.

CLINICAL CORRELATE

Myasthenia Gravis

Myasthenia gravis is an autoimmune, neuromuscular disease that affects the acetylcholine receptors of the neuromuscular junction. In this disease, autoantibodies are produced that block or damage the **acetylcholine receptor** of the motor endplate, inhibiting the normal effects of acetylcholine at these receptors and thus producing muscle weakness. Muscles of the eye and face and muscles involved in swallowing, talking, and chewing are most commonly affected, although weakness of other muscles may also occur. Episodes of myasthenia gravis may occur suddenly, often after a period of high physical activity, and may abate after a period of

rest. A myasthenic crisis is a medical emergency that may be caused by an infectious disease or adverse drug reaction. In a crisis, muscles of respiration are weakened, making breathing difficult. Such cases require assisted ventilation, such as positive airway pressure ventilation. In myasthenia gravis, the appearance of symptoms is usually episodic, and treatment often involves an immunosuppressive drug such as corticosteroids, as well as a **cholinesterase inhibitor** (for example, neostigmine). Cholinesterase inhibitors inhibit acetylcholinesterase, prolonging the half-life of acetylcholine in the neuromuscular junction. In some cases, plasmapheresis may be used to remove autoantibodies or thymectomy may be performed.



Clinical Manifestations of Myasthenia Gravis In myasthenia gravis, an autoimmune disease, antibodies block or reduce the number of nicotinic acetylcholine receptors at the neuromuscular junction, resulting in muscle fatigability. Diagnostic tests include the edrophonium test. Edrophonium chloride is a cholinesterase inhibitor and thus increases acetylcholine at the neuromuscular junction. Administered intravenously, it will temporarily relieve symptoms of muscle weakness, including diplopia (double vision), in myasthenia gravis.

THE SLIDING FILAMENT THEORY

The sliding filament theory describes the processes that take place in the sarcomeres that account for skeletal muscle contraction. The sliding of the interdigitated, anchored thick and thin filaments is the basis of contraction (Fig. 3.14). The thick filaments of the sarcomeres are composed of the protein **myosin** and are anchored to the **M line**; the thin fila-

ments are composed of **actin**, **tropomyosin**, and **troponin** and are anchored to the **Z line** (Fig. 3.15). Within the thin filament, globular G-actin is polymerized to form an α -helical filament (F-actin). Actin has binding sites for myosin along the groove of its helix; these sites are covered by the protein **tropomyosin**. Three forms of **troponin** are also incorporated at regular intervals (troponin C, troponin I, and troponin T).

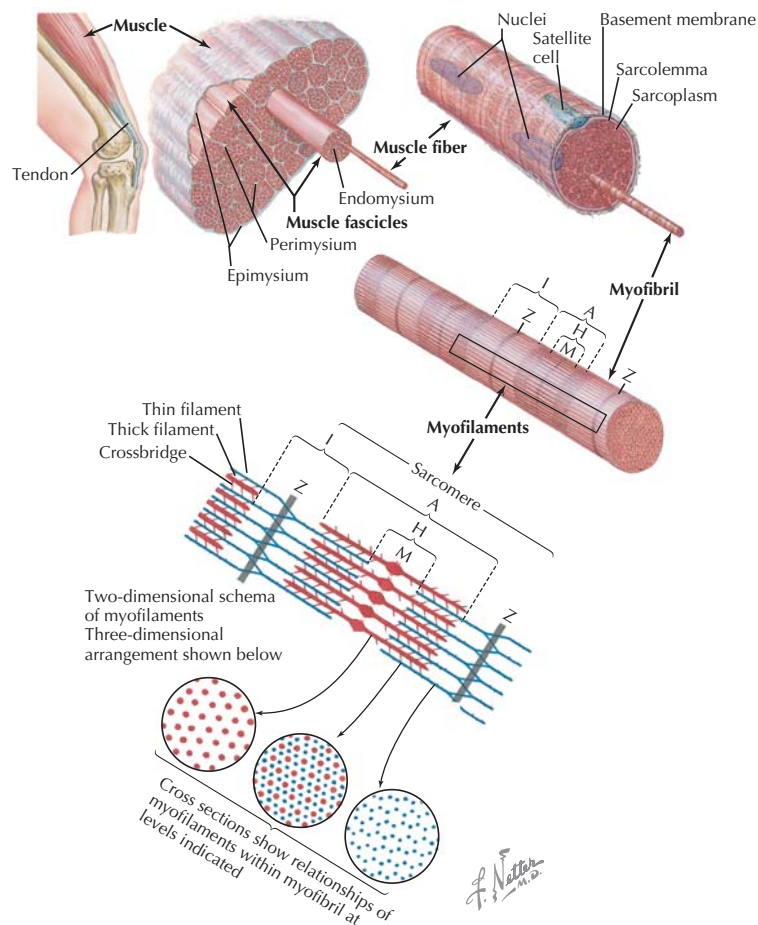


Figure 3.11 Organization of Skeletal Muscle Skeletal muscle is composed of fascicles that are in turn comprised of multinucleated muscle fibers. These fibers are composed of smaller myofibrils, which contain sarcomeres, the site at which sliding of actin and myosin filaments produces contraction. The organization of sarcomeres within the skeletal muscle produces its striated appearance. The Z line marks the boundary between two sarcomeres. The I band contains only the actin thin filaments, which extend from the Z line toward the center of the sarcomere. Myosin thick filaments are found in the dark A band. At the H zone, there is no overlap between actin and myosin. The M line is at the center of the sarcomere and is the site at which the thick filaments are linked with each other.

In the relaxed state, in which cytosolic Ca^{2+} concentration is extremely low (10^{-7} M), binding of myosin to actin is blocked because myosin binding sites on actin are covered by troponin. Partially hydrolyzed ATP (ADP) is bound to the myosin head groups (see Fig. 3.15). When an action potential causes release of Ca^{2+} from the sarcoplasmic reticulum, binding of Ca^{2+} to troponin causes troponin to move into the groove of the actin α -helix. This results in exposure of the myosin binding sites, thus promoting the binding of the myosin head group to actin, forming a **crossbridge**. This binding is followed by a ratcheting action of the myosin head group, shortening the sarcomere as the actin and myosin slide past each other. ADP and inorganic phosphate (P_i) are released. Next, binding of ATP to the myosin head group causes detachment of actin and myosin, after which ATP is partially hydrolyzed



Skeletal muscle fibers can be categorized as **fast twitch** and **slow twitch** fibers, also known as white muscle and red muscle, respectively. Most muscles in the body are composed of a mixture of the two types, but individual motor units contain only one fiber type. The oxygen-binding protein **myoglobin** is found only in slow twitch fibers and is responsible for its red color. Fast twitch (white muscle) fibers utilize anaerobic glycolysis as an energy source and therefore are high in glycogen content. This type of muscle is utilized for rapid, powerful movements; ATPase activity is high and contraction speed is high, and resistance to fatigue is low. Slow twitch (red muscle) fibers utilize oxidative phosphorylation, and thus glycogen content is low. Slow twitch fibers are used for activities requiring endurance. ATPase activity and contraction speed are low, and resistance to fatigue is high.

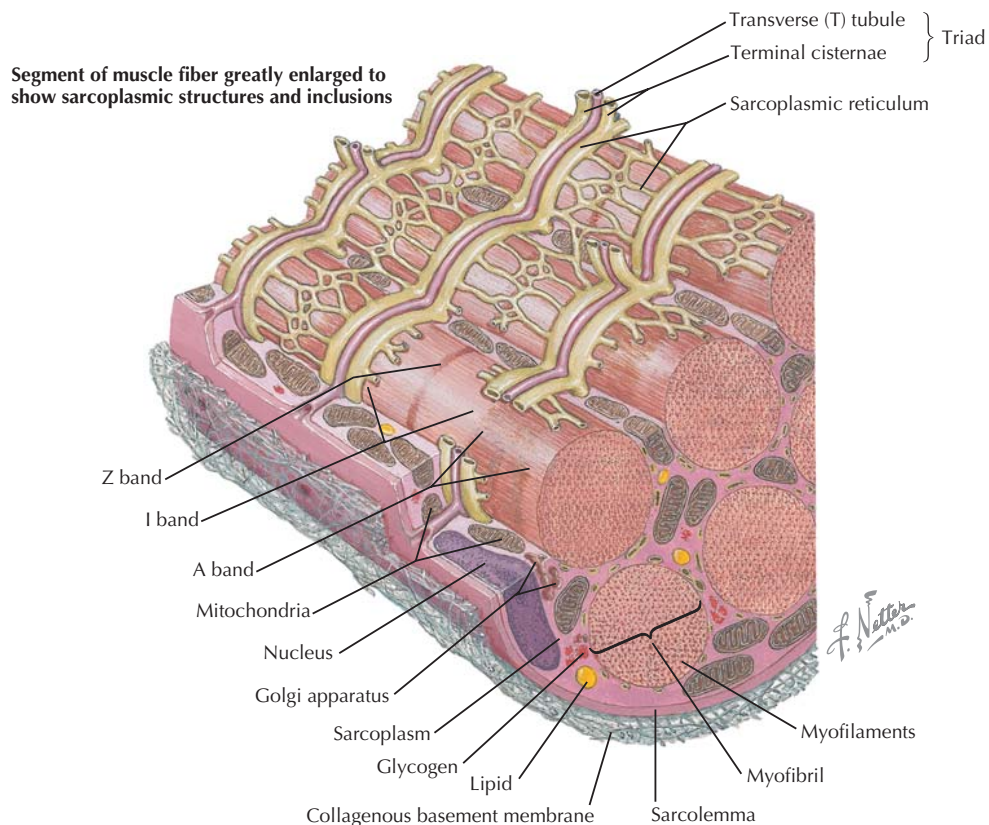


Figure 3.12 Sarcoplasmic Reticulum The sarcoplasmic reticulum is a complex network surrounding the myofibrils and storing high concentrations of Ca^{2+} , sequestered from the sarcoplasm. The membrane of the sarcoplasmic reticulum contains Ca^{2+} -ATPase, which is essential for this sequestration. The transverse tubules are deep invaginations of the sarcolemma and form triads with the terminal cisternae of the sarcoplasmic reticulum. These transverse tubules conduct the action potential from the sarcolemma to the cisternae, causing release of Ca^{2+} .

by ATPase; this causes “recocking” of the head group. If cytosolic Ca^{2+} is still elevated, myosin rapidly binds to actin, and crossbridge cycling in this manner causes the contraction to continue. Crossbridge cycling occurs simultaneously at multiple sites, producing the shortening of the sarcomere. When Ca^{2+} is resequenced into the sarcoplasmic reticulum, muscle relaxes due to low cytosolic Ca^{2+} concentration.

During crossbridge cycling, each cycle requires one ATP molecule to dissociate each myosin head group. Furthermore, relaxation requires ATP, for sequestration of Ca^{2+} into the sarcoplasmic reticulum. In **rigor mortis**, because ATP is depleted, myosin remains bound to actin, causing stiffness of the muscles.

MECHANICAL CONSIDERATIONS IN SKELETAL MUSCLE CONTRACTION

Generation of force by skeletal muscle is controlled by a number of factors (Fig. 3.16). For example, variation in the

size of motor units is consistent with the function of the motor units. Small motor units are effective in performing fine movements, as in movements of the fingers and eyes, whereas large motor units perform coarser movements (see Fig. 3.16A). The force of contraction of skeletal muscles can be increased by **recruitment** of motor units and **summation of twitches**. As noted earlier, all of the muscle fibers associated with a single motor unit will contract simultaneously. Greater contraction can be achieved by recruiting more motor units (**spatial summation**). With repeated stimulation of a muscle, **temporal summation** may occur. In this case, individual twitches of the muscle occur without full relaxation between twitches, with greater force of contraction resulting as Ca^{2+} is released from the sarcoplasmic reticulum at a higher rate than it can be resequenced between the twitches (see Fig. 3.16B). In **tetanus** (tetany), there is a sustained, forceful contraction due to high frequency stimulation.

Finally, the muscle tension generated during contraction is dependent on the resting tension or the degree of stretch of the muscle at rest. In other words, there is a **length–tension**

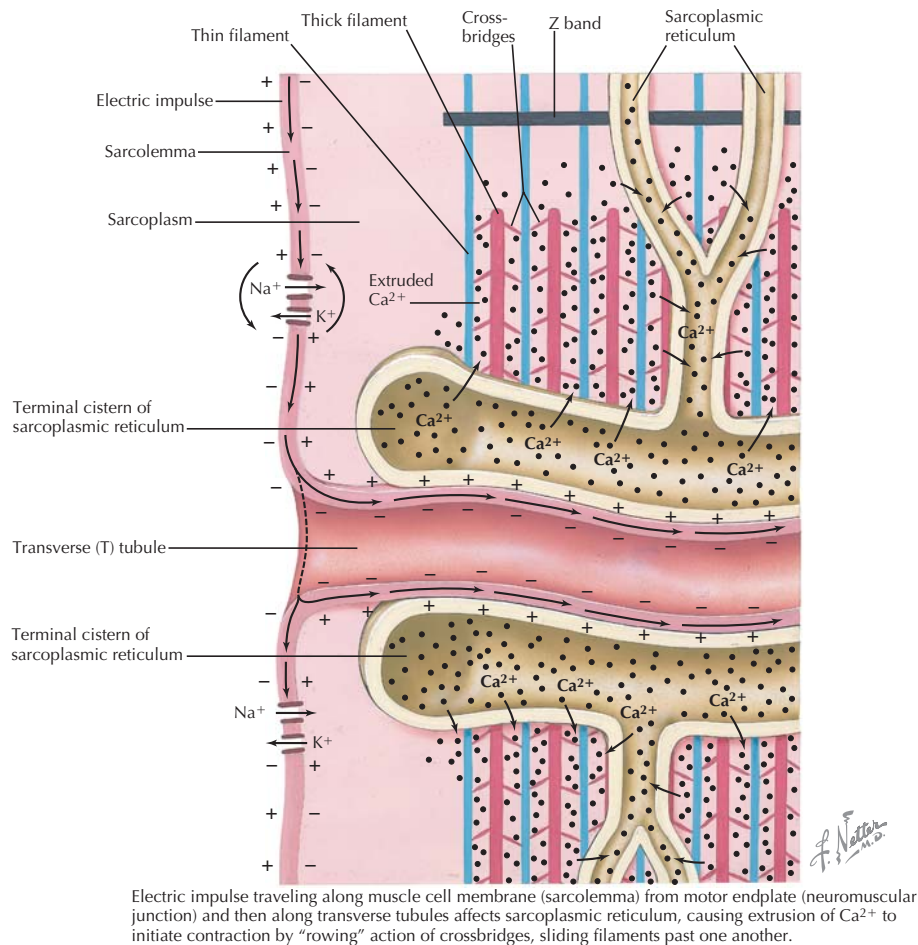


Figure 3.13 Excitation-Contraction Coupling When an action potential in a motor neuron results in the release of acetylcholine at the neuromuscular junction, binding of acetylcholine on the sarcolemma opens a cation channel, permitting influx of Na^+ . An action potential is produced and spreads into the transverse tubules, resulting in the release of Ca^{2+} stored in the sarcoplasmic reticulum, initiating crossbridge formation and muscle contraction. Ca^{2+} is resequstrered into the sarcoplasmic reticulum by Ca^{2+} -ATPase to terminate the contraction.

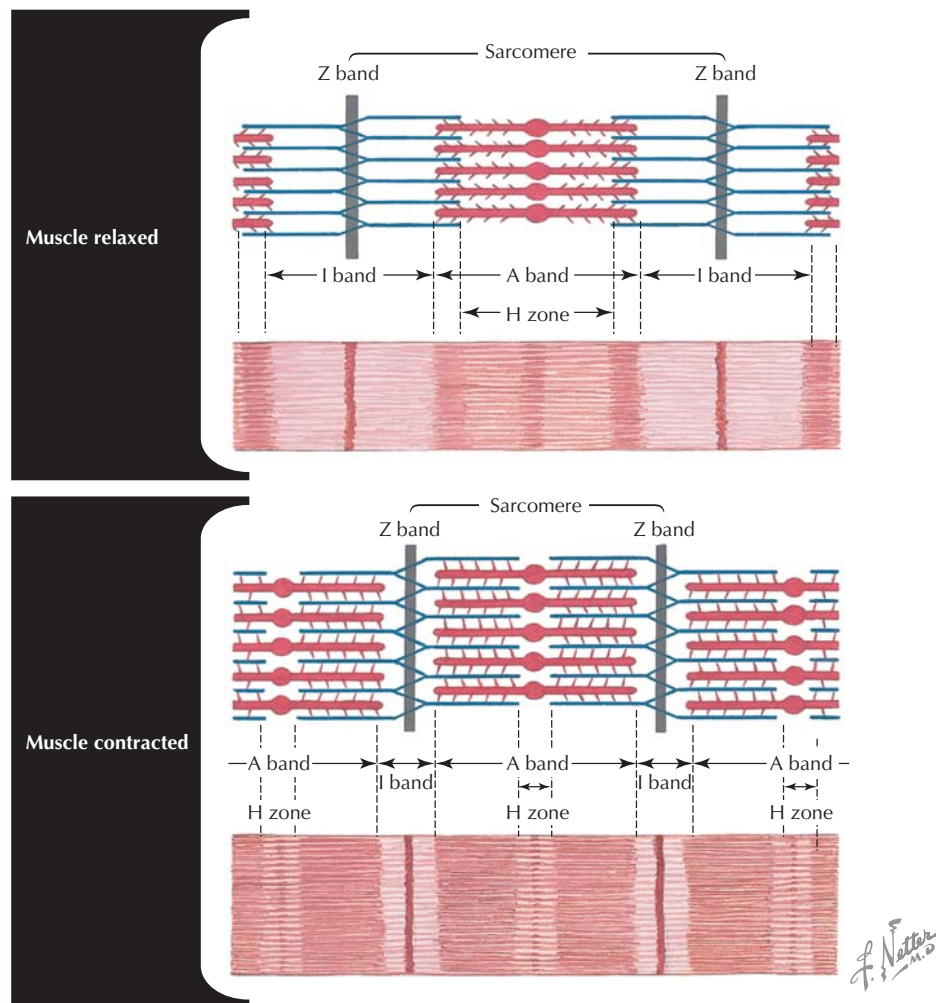
relationship for skeletal muscle (see Fig. 3.16C). The extent of overlap of the thin and thick filaments of the sarcomeres affects the number of crossbridges formed during contraction. With greater stretch before stimulation, more force is generated, up to an optimal sarcomere length.

Muscle contractions can also be characterized as isometric or isotonic. In **isometric contraction**, muscle length is constant but a change in tone is observed. An example is the contraction that occurs when we exert force against an immovable object. Force is generated, but length remains unchanged. Sarcomere shortening is accompanied by stretch of connective tissue and cytoskeletal components of the muscle (**series elastic elements**). During **isotonic contraction**, muscle tone remains constant while the muscle shortens. An example is the contraction associated with lifting a book. In reality, all muscle movements are a combination of isometric and iso-

tonic contractions. Pressing of heavy free weights involves a large isometric component, but it also involves an isotonic component. In general, isometric exercises like weightlifting are useful for body-building. With repeated isometric exercise, **hypertrophy** of muscle occurs, in which the number of sarcomeres is increased (**hyperplasia**, or increased number of muscle cells, does not normally occur). Muscle **atrophy** or wasting (decreased number of sarcomeres) occurs with disuse, for example during prolonged bed rest.

SMOOTH MUSCLE

Smooth muscle is a type of nonstriated muscle found within organs. Contractile proteins are not organized as sarcomeres in smooth muscle; rather, actin is anchored to the cell membrane and to dense bodies within the cell. As in other types of



During muscle contraction, thin filaments of each myofibril slide deeply between thick filaments, bringing Z bands closer together and shortening sarcomeres. A bands remain the same width, but I bands narrow. H zones also narrow or disappear as thin filaments encroach upon them. Myofibrils and, consequently, muscle fibers (muscle cells), fascicles, and muscle as a whole grow thicker. During relaxation, the reverse occurs.

Figure 3.14 Muscle Contraction and Relaxation During skeletal muscle contraction, cyclical crossbridge formation between interdigitated actin and myosin filaments produces shortening of the sarcomere.

muscle, actin–myosin interactions are the basis of contraction (Fig. 3.17). A comparison of skeletal, smooth, and cardiac muscle is shown in Table 3.3.

Types of Smooth Muscle

Smooth muscle is further classified as unitary or multiunit types. Of these types, **unitary smooth muscle** is far more abundant and is found in the walls of blood vessels, the bladder and the gut, among other organs. This type of smooth muscle is capable of sustained and often powerful contractions. The cells have gap junctions between them, allowing rapid and direct spread of action potentials, as electrical potentials are directly conducted between cells

through ion fluxes. Because of this, many smooth muscle cells act as a single unit, producing a synchronous contraction. In contrast, **multiunit smooth muscle** is organized into motor units similar to those in skeletal muscle. Cells are electrically isolated from each other (there are no gap junctions), allowing for fine motor control. This type of muscle is found in a few specific regions such as the ciliary body of the eye, the vas deferens, and the piloerector muscles in the skin.

Contraction of Smooth Muscle

Smooth muscle contraction is controlled by multiple neurotransmitters and other chemical ligands that affect cytosolic